

The ordinary-double-point gap in every dimension

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Abstract

We prove the ordinary-double-point gap conjecture for boundary-zero complex algebraic klt germs in every dimension: a singular n -dimensional germ has normalized volume at most $2(n-1)^n$, with equality precisely at an analytic ordinary double point.

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1 Introduction

Normalized volume, introduced by Chi Li [26], measures the size of a klt singularity by balancing the discrepancy of a valuation against the asymptotic colength of its valuation ideals. For a closed point x of an n -dimensional normal complex variety that is klt near x , with zero boundary, it is

$$\widehat{\text{vol}}(x, X) = \inf_{v \text{ centered at } x} A_X(v)^n \text{vol}(v).$$

Here $A_X(v)$ is log discrepancy, and $\text{vol}(v)$ is the leading colength of the ideals $\{f : v(f) \geq t\}$, multiplied by $n!$. Section 2 gives the precise conventions for real valuations, including infinite discrepancy. In dimension n , the largest normalized volume is n^n , attained exactly at smooth points [36, Theorem 1.6]. The ordinary double point gap conjecture predicts the next largest value: every singular klt germ should have normalized volume at most $2(n-1)^n$, and this value should characterize the ordinary double point. Besides distinguishing a basic singularity by a numerical invariant, such a gap gives global restrictions on K-semistable Fano varieties through the local-to-global volume inequality [33, Theorem 21].

An ordinary double point is the analytic hypersurface germ of a nondegenerate quadratic form. It is the simplest singular quadratic cone, and the asserted gap concerns all singular klt germs, not only hypersurfaces. Thus an essential part of the problem is to extract structural information from a numerical lower bound on volume.

Theorem 1.1 (The ordinary double point gap). *Let $n \geq 2$, and let x be a singular closed point of a normal complex algebraic n -fold X . Suppose that K_X is $\mathbb{Q}$ -Cartier and that X is klt near x , with zero boundary. Then*

$$\widehat{\text{vol}}(x, X) \leq 2(n-1)^n.$$

Equality holds if and only if

$$(X, x)_{\text{an}} \cong (\{z_0^2 + \cdots + z_n^2 = 0\} \subset \mathbb{C}^{n+1}, 0).$$

The point need not be isolated in the singular locus. No structural assumption on the singularity is implicit in Theorem 1.1. We use log discrepancy, so the discrepancy of the exceptional divisor of the blowup of a smooth n -fold point is n . The normalized volume at the ordinary double point is $2(n-1)^n$, including its interpretation as an infimum over all centered valuations.

The proof uses the fourfold theorem of the companion paper *The normalized-volume gap in dimension four* [44, Theorem 1.1] as its sole companion input. That theorem has the same unrestricted boundary-zero scope and includes the analytic equality characterization; we restate it in Section 2. The induction from this base case, developed below, proves both the bound and the equality assertion in every higher dimension.

Corollary 1.2 (Global volume bounds). *Let V be an n -dimensional K -semistable $\mathbb{Q}$ -Fano variety, where $n \geq 2$. If V is singular, then*

$$(-K_V)^n \leq 2 \left(\frac{n^2 - 1}{n} \right)^n < 2n^n.$$

If $V \not\cong \mathbb{P}^n$, then $(-K_V)^n \leq 2n^n$, with equality precisely when V is a smooth quadric $Q^n \subset \mathbb{P}^{n+1}$ or $\mathbb{P}^1 \times \mathbb{P}^{n-1}$. These two descriptions coincide for $n = 2$.

Here a $\mathbb{Q}$ -Fano variety is normal, projective and klt, with ample $\mathbb{Q}$ -Cartier anticanonical divisor. The smaller singular bound in Corollary 1.2 is not accompanied by an assertion of sharpness or a classification of its equality cases.

1.1 Earlier results

The gap conjecture arose from the study of singular noncollapsed polarized Kähler–Einstein limits, where the volume density measures the size of a metric tangent cone. The algebraicity and uniqueness of these tangent cones were established by Donaldson–Sun [10]. Spotti–Sun proposed that the ordinary double point has the largest density among singular Ricci-flat Kähler cones with isolated vertex, and formulated the corresponding normalized-volume conjecture for arbitrary singular klt germs [46, Conjectures 1.2 and 5.5]. Their observation that density above $1/2$ forces Gorenstein canonical structure in the metric-limit setting [46, Proposition 3.7] already illustrates a central feature of the problem: a sufficiently large volume can force algebraic structure that is absent from the hypotheses. Here that principle is applied after stable degeneration, starting from an arbitrary klt germ.

The unrestricted low-dimensional cases precede the present induction. For surfaces, Li–Liu’s calculation $\widehat{\text{vol}}(0, \mathbb{C}^2/G) = 4/|G|$ for finite groups acting freely in codimension one, together with the quotient classification, gives the bound and its equality case [27, Proposition 4.10]. Liu–Xu proved the threefold theorem for arbitrary klt germs [36, Theorem 1.3]. In higher dimensions, Liu proved the conjecture for local complete intersections [34, Theorem 1.3]. For singular $\mathbb{Q}$ -Gorenstein toric germs, Moraga–Süß proved the different

sharp bound $\widehat{\text{vol}} \leq (16/27)n^n$ for $n \geq 3$, with equality precisely for the product of a three-dimensional ordinary double point and $\mathbb{C}^{n-3}$; in dimension two the bound is 2, with equality at the surface ordinary double point [41, Theorem 2]. The fourfold companion supplies the remaining base case used here; the higher-dimensional argument does not assume either a complete-intersection or a toric structure.

The global application combines the local gap with a complementary smooth theorem. Li–Miao proved the bound $2n^n$ and its equality classification for K-semistable Fano manifolds other than projective space [28, Theorem 1.1]. Li–Miao–Zhang established the corresponding metric gap for compact Kähler manifolds other than projective space with $\text{Ric}(\omega) \geq (n+1)\omega$, and a global degree bound for K-semistable toric log Fano pairs whose underlying variety is not projective space [32, Theorems 1.3 and 1.6]. The local theorem here gives a strict improvement for singular K-semistable $\mathbb{Q}$ -Fano varieties, so that equality at $2n^n$ reduces to Li–Miao’s smooth classification. Further local refinements address the range below the threefold maximum: Liu proves the bound 9 for Gorenstein canonical non-hypersurface threefolds and classifies all klt threefold germs with normalized volume at least 9 [35, Theorems 1.1 and 1.2].

The passage from metric cones to arbitrary klt singularities rests on the variational theory of normalized volume. Martelli–Sparks–Yau proved volume minimization among Reeb fields with fixed canonical weight [39]; Li–Liu and Li–Xu related cone metrics and K-stability to minimization over all centered valuations [27, Theorem 1.9][29, Theorem 1.3]. For general klt germs, existence of a minimizer, its quasi-monomiality, uniqueness up to scaling, and finite generation were established by Blum, Xu, and Xu–Zhuang [4, 50, 51, 52]. These results produce a K-semistable cone whose Reeb valuation computes the original normalized volume. The further degeneration of Li–Wang–Xu gives a K-polystable cone with the same value [31, Theorem 1.2]. Thus the cone reduction preserves the infimum over all valuations, which is essential for a gap theorem about arbitrary germs.

We also use lower semicontinuity [5] and the finite degree formula [51] to derive smoothness of the puncture and Gorenstein structure from the density threshold. The new argument begins after that reduction: it must control integral gradings approximating the minimizing Reeb vector of an isolated Gorenstein cone, whose quotient is usually an orbifold. Passing to the coarse quotient too early loses the cyclic characters that control both rational curves and the eventual volume estimate.

Several classical methods enter at this point, in different roles. The use of rational curves belongs to the bend-and-break approach initiated by Mori [42], with stack-theoretic extensions such as Chen–Tseng [6]. The particular curve needed here has at most one stack point and comes from Li–Zhou [30]. Properness of twisted stable maps [1] turns its minimal degree into a finite evaluation map. The Ricci-flat cone metric supplied by Collins–Székelyhidi [8] then bounds the weights of symmetric tensors obtained from this map. This is related in method to Reeb-weight obstructions for holomorphic functions [13], but the symmetric-tensor estimate is proved here. The final high-index classification is the projective-space and quadric characterization of Kobayashi–Ochiai [22]; the needed ample-line-bundle formulation also follows from Araujo–Druel–Kovács [2].

The other case uses an adjoint construction. Howald’s multiplier-ideal formula [17] and Kawamata–Viehweg vanishing [12] lift a prescribed leading Taylor monomial after the canonical shift. Hochster’s theorem and Stanley’s canonical-module formula [16, 47] subsequently convert the resulting normal semigroup into a Gorenstein degeneration. None of those statements alone forces the canonical point to be interior: that separate numerical assertion is the content of the exponential estimate in Section 4. Its semigroup counting follows the convex-geometric approach to graded linear series developed by Khovanskii, Okounkov, Lazarsfeld–Mustață, and Kaveh–Khovanskii [21, 43, 25, 19]. The new exact saturation conclusion is stronger than the asymptotic count supplied by that method and

requires the adjoint argument.

1.2 The argument and its additional statements

Put $n = N + 1$ and

$$p_n = 2 \left(\frac{n-1}{n} \right)^n, \quad d(x, X) = \frac{\widehat{\text{vol}}(x, X)}{n^n}.$$

Starting from the known cases in dimensions two and three and the fourfold companion, we argue by induction. In the step $n \geq 5$, for a singular germ with $d(x, X) \geq p_n$, stable degeneration preserves this density and bounds the original embedding dimension by that of a K-polystable cone C . The lower-dimensional gap and the finite degree formula force C to be smooth off its vertex and Gorenstein. These are conclusions of the reduction, rather than assumptions on the germ.

A Reeb vector is a real torus grading positive on every nonconstant homogeneous function. Approximate the minimizing Reeb ray of C by primitive integral gradings. For one such grading, let r be the weight of a generating canonical form and let $m_{\max}$ be the largest stabilizer order on the puncture. The quotient by the grading action is a smooth orbifold B , with faithful ample weight-one line bundle L and $-K_B = rL$. The comparison of $m_{\max}$ with r/N separates the two parts of the argument.

When $m_{\max} \leq r/N$, a minimal orbifold rational curve constrains the tangent characters. Unless those constraints already force the grading to be free, its deformations give a finite evaluation map onto C . Taking the product of their infinitesimal translations over the generic sheets produces a nonzero symmetric tensor. A weight bound from the Ricci-flat cone metric then forces a free grading with canonical weight N or $N + 1$. The high-index classification identifies (B, L) as a quadric or projective space with its standard polarization, so C is an ODP or is smooth.

For $m_{\max} \geq r/N$, choose a point with maximal stabilizer and record each section's degree and leading transverse Taylor exponent. These pairs form a semigroup Γ in the lattice determined by the stabilizer characters; write K for its closed convex cone. The canonical form determines the lattice point $c_0 = (r, 1, \dots, 1)$. Two independent arguments give exact control of Γ . Near-minimality of the Reeb grading gives moment inequalities, and an exponential expectation bound puts c_0 in $\text{int } K$. Adjoint vanishing lifts Taylor monomials whose indices, after the shift by c_0 , lie in that interior. Together these facts force rational polyhedrality and exact saturation of Γ .

Two filtrations convert this semigroup structure into a singular klt slice Y of the same dimension n . A nontrivial stabilizer of order $h \geq 2$ for the second filtration action makes one character account for only $1/h$ of the leading valuation colength on Y . This improves the volume comparison with C . The action and its order h are different from those defining $m_{\max}$; the quadratic-cone example in Section 6 makes the distinction explicit.

To use this same-dimensional slice without assuming the desired bound for it, let M_n be the supremum of densities of singular boundary-zero klt n -fold germs. The comparison gives

$$d(o, C) \leq \frac{p_n + M_n}{2}.$$

Applying this estimate to densities approaching M_n proves $M_n \leq p_n$. The approximation of gradings is made separately on each fixed cone; the supremum need not be attained. In the equality case, a strict colength count and integral rounding return the cone to the small-stabilizer case. Its ODP embedding dimension then makes the original singular germ a hypersurface, and a weighted test forces a nondegenerate quadratic part. Figure 1 records this case split and the return in equality.

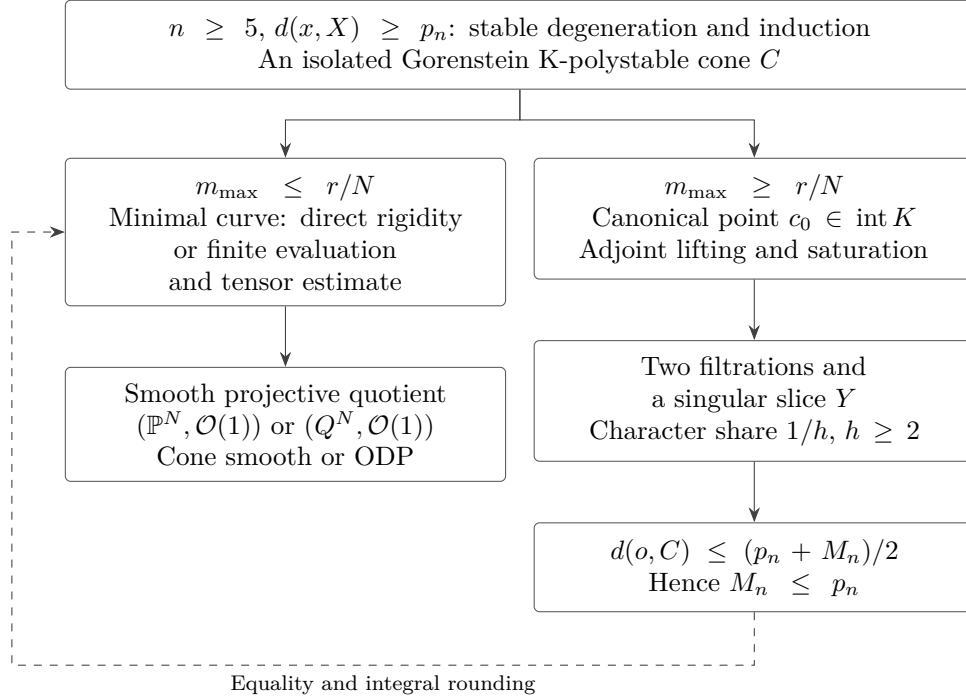


Figure 1: The induction step $n = N + 1 \geq 5$ for a singular germ of density at least p_n . Here K is the cone of leading Taylor indices and c_0 its canonical lattice point. The same-dimensional slice enters only through the supremum M_n of singular n -dimensional densities. Equality in the volume estimate returns the cone to the small stabilizer case.

Organization. Section 2 fixes the valuation conventions, supplies the cone reduction, and proves the absolute grading approximation. Section 3 treats small stabilizers. For the other case, Section 4 proves canonical interiority and Section 5 independently establishes adjoint lifting before combining the two results. Section 6 constructs the two filtrations and the second stabilizer. Section 7 constructs the same-dimensional singular slice, proves the supremum inequality, and analyzes equality. Finally, Section 8 returns to the original germ and derives the Fano consequences.

2 Normalized volume and the cone reduction

2.1 Conventions and public inputs

All varieties and stacks in the proof are over $\mathbb{C}$. Boundaries are zero unless a boundary is explicitly introduced in a vanishing argument. For a klt germ $x \in X$, write $R = \mathcal{O}_{X,x}$ and $\mathfrak{m} = \mathfrak{m}_x$. If E is a prime divisor on a smooth proper birational model $\mu: X' \rightarrow X$, its log discrepancy is

$$A_X(E) = 1 + \text{coeff}_E(K_{X'} - \mu^*K_X).$$

For a real valuation v of $K(X)$, trivial on $\mathbb{C}$ and centered exactly at x , put

$$\mathfrak{a}_t(v) = \{f \in R : v(f) \geq t\}, \quad \text{vol}(v) = \lim_{t \rightarrow \infty} \frac{n! \text{length}(R/\mathfrak{a}_t(v))}{t^n}.$$

The ideals are $\mathfrak{m}$ -primary. The limit exists by [9, Theorem 5.8]: the local rings here are analytically unramified and equicharacteristic with perfect residue field. The integer-cutoff ideals form a graded family, and monotonicity extends the integer limit to real cutoffs.

We use the standard extension of log discrepancy to real valuations. On a smooth birational log model it is linear on each monomial cone; for an arbitrary valuation it is the supremum over its monomial retractions [18, Proposition 5.1 and Corollary 5.4]. For the singular $\mathbb{Q}$ -Gorenstein base one includes the relative canonical correction on a resolution, as in [26, Section 2.1]. Define

$$\widehat{\text{vol}}_X(v) = \begin{cases} A_X(v)^n \text{vol}(v), & A_X(v) < +\infty, \\ +\infty, & A_X(v) = +\infty. \end{cases}$$

Thus the definition never requires multiplying infinity by zero. Our normalization is

$$\widehat{\text{vol}}(x, X) = \inf_v \widehat{\text{vol}}_X(v), \quad d(x, X) = \frac{\widehat{\text{vol}}(x, X)}{n^n}. \quad (2.1)$$

The infimum is unchanged if restricted to positive multiples of divisorial valuations [29, Theorem 2.5]. Rescaling a valuation does not change its normalized volume.

We use the following established facts in their boundary-zero form.

- (i) A klt germ has a minimizing valuation, unique up to positive scaling. It has finite discrepancy, is quasi-monomial, and has finitely generated associated graded. The latter is a K -semistable Fano cone, which admits a special equivariant degeneration to a K -polystable Fano cone. The minimizing Reeb vector and normalized volume are preserved [52, Theorem 1.2], [31, Theorem 1.2]. Uniqueness also implies that a minimizer normalized by discrepancy is invariant under every automorphism of the germ [51, Theorem 1.1 and Corollary 1.2].
- (ii) Normalized volume is lower semicontinuous in $\mathbb{Q}$ -Gorenstein flat families of klt germs with section over a normal base [5, Theorem 1]. In particular it is lower semicontinuous when the point moves along a curve in a fixed klt variety [5, Theorem 33(1)]. One has $d(x, X) \leq 1$, with equality exactly at smooth points [36, Theorem 1.6].
- (iii) For a finite Galois morphism of klt germs $f: (Y, y) \rightarrow (X, x)$, with $f^{-1}(x) = \{y\}$ set-theoretically and $K_Y = f^*K_X$, the finite degree formula is

$$\widehat{\text{vol}}(y, Y) = \deg(f) \widehat{\text{vol}}(x, X). \quad (2.2)$$

We use this only for a quotient by a group without reflections and for the cyclic canonical index cover [51, Theorem 1.3].

- (iv) Isomorphic completed local rings have the same normalized volume [34, Lemma 2.15]. Thus étale germs at complex closed points have the same invariant.

These facts concern arbitrary klt germs. In particular, the first item does not require a divisorial minimizer, isolatedness, or smoothability.

2.2 Products and embedding dimension

Lemma 2.1 (A smooth factor). *Let $y \in Y$ be an N -dimensional klt germ, where $N \geq 1$. Then*

$$d((y, 0), Y \times \mathbb{A}^1) \leq d(y, Y).$$

Proof. Take a divisorial test v centered at y , of discrepancy $A > 0$ and volume σ , and let $a = A/N$. The Gauss extension to the added parameter t is

$$(v \oplus a) \left(\sum_i f_i t^i \right) = \min_i \{v(f_i) + ia\}.$$

On a log resolution this is a monomial valuation along the divisor of v and $t = 0$, so its log discrepancy is $A + a$. Completion does not change the finite colengths. In the power-series ring the quotient decomposes by powers of t , giving

$$\text{length}(R[[t]]/\mathfrak{a}_k(v \oplus a)) = \sum_{0 \leq i < k/a} \text{length}(R/\mathfrak{a}_{k-ia}(v)).$$

The defining asymptotic for σ , followed by a Riemann sum, gives $\text{vol}(v \oplus a) = \sigma/a$. Therefore

$$\frac{(A + A/N)^{N+1}}{(N+1)^{N+1}} \frac{N\sigma}{A} = \frac{A^N \sigma}{N^N}.$$

Take the infimum over the divisorial tests on Y . □

Lemma 2.2 (Lifting generators). *Let v be a finite-discrepancy valuation of a klt germ, with finitely generated positively graded associated graded ring $\text{gr}_v R$. Then*

$$\text{edim}(R) \leq \text{edim}(\text{gr}_v R).$$

Proof. Choose minimal homogeneous algebra generators $\bar{f}_1, \dots, \bar{f}_e$ of $\text{gr}_v R$, all of positive value, and lift them to $f_1, \dots, f_e \in \mathfrak{m}$. To show that their classes span $\mathfrak{m}/\mathfrak{m}^2$, start with $g \in \mathfrak{m}$. Its initial form is a homogeneous polynomial in the $\bar{f}_i$. Subtracting the corresponding polynomial in the f_i either annihilates the remainder or strictly increases its value. Every positive value that occurs belongs to the semigroup generated by the positive numbers $v(f_i)$, so only finitely many such values lie below any fixed bound.

The Izumi inequality for a finite-discrepancy valuation on a klt germ [26, Theorem 3.1] gives $v(g) \leq c \text{ord}_{\mathfrak{m}}(g)$, with a fixed constant c . In particular, a nonzero remainder of value greater than c lies in $\mathfrak{m}^2$. The cancellation process therefore terminates modulo $\mathfrak{m}^2$ after finitely many steps. Every polynomial subtracted has no constant term, and modulo $\mathfrak{m}^2$ is a linear combination of the f_i . This proves the assertion. □

2.3 The minimum at the ordinary double point

Lemma 2.3 (The ODP minimum). *For $n \geq 2$, the ordinary double point Q_n has*

$$\widehat{\text{vol}}(0, Q_n) = 2(n-1)^n.$$

Proof. First take the exact quadratic cone

$$Q_n = \text{Spec } R, \quad R = \mathbb{C}[z_0, \dots, z_n]/(z_0^2 + \dots + z_n^2),$$

with its ordinary grading. It is klt. For example, blowing up the vertex resolves it with smooth exceptional quadric, whose log discrepancy is $n-1 > 0$; further smooth blowups preserve the klt condition.

Normalize a minimizing valuation v by its discrepancy. Uniqueness makes it invariant under all automorphisms, in particular under cone scalings and the connected orthogonal group $\text{SO}(n+1, \mathbb{C})$. For each $j \geq 0$, the degree- j piece R_j is the irreducible representation of complex harmonic polynomials of degree j in $n+1$ variables. To identify the quotient, write P_j for homogeneous complex polynomials of degree j and $\mathcal{H}_j$ for the subspace killed by $\sum_i \partial_{z_i}^2$; set $P_k = 0$ for $k < 0$. For $q = \sum z_i^2$, the harmonic decomposition $P_j = \mathcal{H}_j \oplus qP_{j-2}$ identifies R_j with $\mathcal{H}_j$; see [14, Example 251 and Theorem 257]. The complex representation is irreducible already under the compact group $\text{SO}(n+1)$, so an invariant subspace under $\text{SO}(n+1, \mathbb{C})$ must be trivial or the whole space. For the classical harmonic-polynomial

formulation and its extensions, see also [38, Section 1]. This includes $n = 2$ and $n = 3$. The exception for two ambient variables is outside the range under consideration.

For each cutoff t , the subspace $\{f \in R_j : v(f) \geq t\}$ is invariant, so it is either zero or all of R_j . Every nonzero element of R_j consequently has a common value a_j . If $0 \neq f \in R_1$, then $f^j \neq 0$, and valuation multiplicativity gives $a_j = ja_1$. Invariance under cone scalings separates homogeneous pieces of a cutoff: a finite Vandermonde combination of scalings isolates each piece. Thus, for $g = \sum_j g_j \neq 0$,

$$v(g) = a_1 \min\{j : g_j \neq 0\}.$$

After localizing at the vertex the same assertion holds, since denominators nonzero at the vertex have value zero. The minimizer is therefore proportional to the ordinary degree valuation.

That valuation is the order along the exceptional quadric of the vertex blowup. Hypersurface adjunction gives discrepancy $n - 1$, and its volume is the hypersurface multiplicity 2. It follows that the infimum equals $2(n - 1)^n$.

Finally, an analytic ordinary double point has the same completed local ring as this exact cone. Completion invariance [34, Lemma 2.15] gives the same normalized volume. $\square$

2.4 The induction setup

The companion paper *The normalized-volume gap in dimension four* [44, Theorem 1.1] proves the following base case. We state the entire interface because its equality assertion is part of the theorem propagated here.

Theorem 2.4 (The fourfold gap). *Let x be a singular closed point of a normal complex algebraic fourfold X . Suppose that K_X is $\mathbb{Q}$ -Cartier and X is klt near x , with zero boundary. Then*

$$\widehat{\text{vol}}(x, X) \leq 162.$$

Equality holds if and only if the complex analytic germ is

$$\left(\{z_0^2 + z_1^2 + z_2^2 + z_3^2 + z_4^2 = 0\}, 0\right).$$

No isolatedness, complete-intersection, quotient, or smoothability assumption is made.

For $k \geq 2$ set

$$p_k = 2 \left(\frac{k-1}{k} \right)^k. \quad (2.3)$$

Let M_k be the supremum of $d(x, X)$ over all singular boundary-zero klt k -fold germs: the varieties are normal and complex algebraic, and their canonical divisors are $\mathbb{Q}$ -Cartier. This supremum is finite, since $M_k \leq 1$. The numbers p_k are strictly increasing: for real $t > 1$,

$$\frac{d}{dt} [t \log(1 - 1/t)] = \log(1 - 1/t) + \frac{1}{t-1} > 0.$$

The last inequality is $\log(1 + u) < u$, with $u = 1/(t - 1)$.

The dimension-two case of Theorem 1.1 follows from the quotient classification of klt surface singularities [7, Proposition 6.11]. The germ is analytically isomorphic to $(\mathbb{A}^2/G, 0)$, where G is a finite effective linear group acting without reflections. Analytic isomorphism identifies the completed local rings, so completion invariance transfers the normalized volume to this algebraic quotient. The finite degree formula gives $d = 1/|G|$. If the germ is singular, $|G| \geq 2$, and equality with $p_2 = 1/2$ forces an order-two action without a fixed hyperplane,

namely $(-1, -1)$. This is the A_1 surface singularity. The threefold case is [36, Theorem 1.3(1)], and the fourfold case is Theorem 2.4.

Henceforth, until the induction is closed, assume Theorem 1.1 in dimensions below n , and fix

$$n = N + 1 \geq 5, \quad d(x, X) \geq p_n, \quad (2.4)$$

with x singular. The only use of M_n will be its definition as a supremum, not an unproved upper bound in dimension n .

Proposition 2.5 (Cone reduction). *Under (2.4), there is a singular K -polystable Fano cone $C = \text{Spec } S$, with vertex o , such that*

$$d(o, C) = d(x, X), \quad \text{edim}(X, x) \leq \text{edim}(C, o).$$

The puncture $C \setminus \{o\}$ is smooth, S is a normal Gorenstein ring, and its canonical module has a homogeneous generator. The cone has zero boundary.

Proof. Apply stable degeneration to a minimizer at x , and then take the K -polystable special degeneration. In the first passage the induced Reeb valuation has the same volume and discrepancy as the original minimizer [29, Lemma 4.10]. In the second, equivariant flatness preserves the dimensions of the character spaces and hence the Reeb volume. The relative canonical character is constant, so its discrepancy is also preserved. Li–Xu’s characterization of K -semistability [29, Theorem 1.3] shows that the final Reeb valuation minimizes normalized volume over all centered valuations. Thus the two passages preserve the infimum in (2.1), not only the number of an arbitrary test.

Lemma 2.2 gives the embedding-dimension inequality for the first passage. Upper semi-continuity along the vertex section of the special flat degeneration gives it for the second. The final vertex is therefore singular.

Choose any positive integral grading of S , and let $P \neq o$. Its grading orbit specializes to o , so lower semicontinuity gives

$$d(o, C) \leq d(P, C). \quad (2.5)$$

Choose a homogeneous function f , of degree $d > 0$, with $c = f(P) \neq 0$, and put $Y_P = \{f = c\} \subset C_f$. The action map

$$Y_P \times \mathbb{G}_m \longrightarrow C_f, \quad (Q, a) \longmapsto a \cdot Q,$$

is the finite étale cover obtained by adjoining a root $a^d = f/c$. Near $(P, 1)$, it expresses the germ as the product of the N -dimensional slice (Y_P, P) and a smooth curve. The slice is klt by this étale description. If P were singular, the slice would be singular as well. The induction hypothesis, Lemma 2.1, and étale invariance would then imply

$$d(P, C) \leq p_N < p_n \leq d(o, C),$$

contrary to (2.5). Thus the puncture is smooth.

Let q be the canonical index at o . The cyclic index cover has degree q , is quasi-étale and crepant, and is klt. It has one point over o . Indeed, its algebra is the direct sum of the reflexive canonical powers modulo q . If a homogeneous element of a nonzero residue degree were not nilpotent in the fiber over o , a power returning to degree zero would be a unit. The corresponding section would trivialize a proper positive canonical power, contradicting minimality of q . Thus the nonzero residue-degree pieces are nilpotent, and the reduced fiber consists of one point. Equation (2.2) and the upper bound $d \leq 1$ give $d(o, C) \leq 1/q$. Since $p_n > 1/2$, necessarily $q = 1$.

Klt singularities are Cohen–Macaulay [24, Definition 117 and Theorem 120]. The canonical module ω_S is invertible at the vertex, and a homogeneous element generating its one-dimensional fiber there generates it after localization. Its graded cokernel has empty support: any nonempty closed support invariant under a positive grading contains the vertex. Since ω_S is torsion-free, the resulting map $S \rightarrow \omega_S$ is injective as well. It is therefore an isomorphism, with a grading shift. $\square$

2.5 Integral gradings and orbifold quotients

Fix the cone of Proposition 2.5. Let $\mathbb{T}$ be an effective algebraic torus of automorphisms containing its Reeb vector ξ , and normalize

$$A(\xi) = n.$$

Choose a $\mathbb{T}$ -eigen-generator Ω_C of ω_S . Its character pairs with cocharacters as the linear discrepancy functional A [29, Lemmas 2.16 and 2.18]. Function weights use the convention $g(aP) = a^j g(P)$; the differential dg has the same weight. The Reeb cone consists of cocharacters positive on all nonconstant homogeneous functions.

Lemma 2.6 (Fast primitive approximations). *There are primitive positive integral cocharacters ζ_i , of canonical weights $r_i = A(\zeta_i) \in \mathbb{Z}_{>0}$, such that*

$$\zeta_i = \frac{r_i}{n}\xi + \epsilon_i, \quad \epsilon_i \longrightarrow 0 \quad (2.6)$$

in the real cocharacter space. If the ray of ξ is rational, the sequence may be chosen constant and exact. Otherwise $r_i \rightarrow \infty$.

The normalized volumes of the degree valuations wt_{ζ_i} tend to $\widehat{\text{vol}}(o, C)$. For any fixed finite set of homogeneous algebra generators, their ζ_i -degrees lie between cr_i and Cr_i for positive constants c, C . Consequently the largest stabilizer order $m_{\max, i}$ on the puncture is $O(r_i)$.

Proof. In the quotient of real cocharacter space by its integral lattice, recurrence gives $t_i \rightarrow \infty$ and integral cocharacters q_i with $q_i = t_i\xi + e_i$, where $e_i \rightarrow 0$. One may obtain such a sequence from simultaneous Dirichlet approximation, or from returns to zero of the one-parameter subgroup in its compact closure. Let d_i be the gcd of q_i in a lattice basis and put $\zeta_i = q_i/d_i$. Then

$$\zeta_i - \frac{A(\zeta_i)}{n}\xi = \frac{1}{d_i} \left(e_i - \frac{A(e_i)}{n}\xi \right) \longrightarrow 0.$$

The cocharacters are positive for large i . If their positive integers $r_i = A(\zeta_i)$ had a bounded subsequence, (2.6) would make ζ_i bounded on that subsequence. A constant lattice subsequence would then give $\zeta = (r/n)\xi$, forcing the ray to be rational. The rational case is handled by its primitive generator.

Choose homogeneous algebra generators $g_1, \dots, g_a$ with $\mathbb{T}$ -characters $\chi_1, \dots, \chi_a$. Each $\langle \chi_j, \xi \rangle$ is positive, so (2.6) gives the asserted two-sided bounds for their integral degrees. More precisely, the normalized cocharacters $(n/r_i)\zeta_i$ converge to ξ . Uniform comparison on the finite set of generator characters then compares their values on every occurring character: each is a sum of generator characters with nonnegative integer coefficients. It follows that the valuation ideals of the normalized degree valuations are squeezed between those of wt_ξ with cutoff factors tending to one. Their volumes therefore converge; their discrepancies are all n .

At a nonvertex point some generator is nonzero. Its stabilizer for a primitive grading is cyclic, and its order divides the degree of every nonvanishing generator. It is therefore at most the largest generator degree, proving the final assertion. $\square$

In what follows, *sufficiently fast gradings* always means gradings in a sequence as in Lemma 2.6, including the exact constant sequence on a rational ray. All constants may depend on the fixed cone. No uniformity over the class defining M_n is asserted or needed.

Lemma 2.7 (The quotient orbifold). *For a primitive positive integral grading ζ , of canonical weight r , let*

$$B = [(C \setminus \{o\})/\mathbb{G}_m].$$

Then B is a smooth effective proper orbifold of dimension N , with projective coarse space $\text{Proj } S$. It has a faithful ample weight-one line bundle L such that

$$S_j = H^0(B, jL), \quad -K_B = rL. \quad (2.7)$$

Every stabilizer is a finite cyclic group. At a point of order m , a smooth transverse uniformizer carries a faithful μ_m -action, and L has a faithful fiber character.

Proof. The smooth puncture has finite stabilizers for the positive grading, so its quotient stack is smooth and Deligne–Mumford. It is the usual stack associated with a positively graded affine ring away from its vertex; its coarse space is $\text{Proj } S$, and a sufficiently divisible positive power of the weight-one line descends to an ample coarse line bundle. This also proves properness.

The grading is effective because the integral cocharacter is primitive in an effective torus. At a point, the finite stabilizer acts trivially on the tangent line of the orbit. If an element acted trivially also on a transverse tangent space, it would have identity differential on the smooth puncture. A finite-order automorphism in characteristic zero is formally linearizable at a fixed point; hence that element would be the identity on a neighborhood and thus on the irreducible cone, contrary to effectiveness. This proves faithfulness of the transverse representation and effectiveness of the orbifold. The weight-one character of the subgroup of $\mathbb{G}_m$ is faithful by construction.

Sections of jL are the homogeneous functions of degree j on the puncture. Normality and the codimension of the vertex extend them to C , giving the first equality in (2.7). Contracting Ω_C with the Euler field gives a horizontal top form of character r , so $K_B = -rL$. $\square$

We retain the stack B , its faithful line L , and the character information throughout the argument. In particular, we do not replace L by a coarse line bundle before the stabilizers have been shown to be trivial.

3 Small stabilizers

We use Proposition 2.5 and Lemmas 2.6 and 2.7. Thus $C = \text{Spec } S$ is a normal Gorenstein Fano cone of dimension $n = N + 1$, where $N \geq 4$, its puncture $C^\circ = C \setminus \{o\}$ is smooth, and the polarized cone (C, ξ) is K-polystable, with minimizing Reeb vector normalized by $A(\xi) = n$. Fix homogeneous torus eigenfunctions $f_1, \dots, f_q$ generating S , with torus characters $\alpha_1, \dots, \alpha_q$. For a primitive positive integral grading ζ , write

$$r = A(\zeta), \quad B = [C^\circ/\mathbb{G}_m], \quad -K_B = rL.$$

Here L is the faithful ample orbifold line bundle of weight one. The maximum stabilizer order for this grading is denoted by $m_{\max}$. All constants in this section may depend on the fixed cone, its torus and the displayed generators.

Theorem 3.1 (Small-stabilizer classification). *Let ζ_i be primitive positive integral gradings satisfying*

$$\zeta_i = \frac{r_i}{n}\xi + \epsilon_i, \quad r_i = A(\zeta_i), \quad \|\epsilon_i\| \rightarrow 0, \quad m_{\max}(\zeta_i) \leq \frac{r_i}{N}. \quad (3.1)$$

For all sufficiently large i , every stabilizer is trivial and $r_i \in \{N, N+1\}$. More precisely, the polarized quotient is

$$(B_i, L_i) \cong \begin{cases} (\mathbb{P}^N, \mathcal{O}_{\mathbb{P}^N}(1)), & r_i = N+1, \\ (Q^N, \mathcal{O}_{Q^N}(1)), & r_i = N, \end{cases}$$

where $Q^N \subset \mathbb{P}^{N+1}$ is a smooth quadric. Consequently the cone C is respectively smooth or an ordinary double point. The assertion includes the constant exact primitive grading when the ray of ξ is rational.

We begin with a minimal orbifold rational curve. Its tangent characters either force $r = N$ and trivial stabilizers, or its deformations yield a finite evaluation map and hence a nonzero symmetric tensor of negative weight. A metric lower bound for homogeneous symmetric tensors, together with an approximation estimate uniform in their rank, gives the remaining rigidity.

3.1 A minimal orbifold rational curve

For the moment fix one primitive grading satisfying

$$m_{\max} \leq r/N. \quad (3.2)$$

The orbifold B is smooth and effective, is proper with projective coarse space, and has ample anticanonical bundle rL . We use the orbifold rational-curve theorem of Li–Zhou [30, Proposition 2.6]. In this setting it supplies a representable nonconstant map from $\mathbb{P}(1, \ell)$, where the only possible stack point is infinity, whose anticanonical degree plus the inverse age at that point is at most $\dim B + 1$. In particular its anticanonical degree is at most n . No positivity assumption on T_B is required for this assertion.

Choose a map of least positive L -degree among all representable nonconstant maps of this form, allowing $\ell = 1$. Such a minimum exists: representability embeds the source stabilizer in a target stabilizer, so the denominators of these degrees divide one of the finitely many integers at most $m_{\max}$. Write

$$f : \mathbb{P}(1, \ell) \rightarrow B, \quad f^*L \cong \mathcal{O}(e), \quad e > 0,$$

and let m be the stabilizer order of its value at infinity. This endpoint order need not equal $m_{\max}$. Thus $\ell \mid m$, and minimality and the existence theorem give $re/\ell \leq n$. On the other hand, $r/m \geq N$ by Equation (3.2). Hence

$$1 \leq \frac{em}{\ell} = \frac{re/\ell}{r/m} \leq \frac{N+1}{N} < 2. \quad (3.3)$$

The expression em/ℓ is an integer. We obtain

$$e = 1, \quad \ell = m, \quad N \leq r/m \leq n. \quad (3.4)$$

Choose an identification $f^*L \cong \mathcal{O}(1)$. Passing to its grading torsor gives an equivariant map

$$F : \mathbb{A}^2 \setminus \{0\} \rightarrow C^\circ, \quad F(tx, t^m y) = t \cdot F(x, y).$$

Since $\mathbb{A}^2$ is normal and C is affine, the coordinate functions extend uniquely across the origin. Thus $F : \mathbb{A}^2(1, m) \rightarrow C$ is a homogeneous polynomial map. Set $P = F(0, 1)$; its stabilizer is μ_m . Conversely, a homogeneous map with no preimage of the vertex outside the origin, and with this degree-one torsor identification, gives a representable map to B : its maps on source inertia groups are the inclusions inherited from the identity homomorphism of the grading groups.

Lemma 3.2 (Deformations with a fixed lift). *Let $b'_1, \dots, b'_n \in \{1, \dots, m\}$ represent the characters of μ_m on $T_{C,P}$, with the trivial character represented by m . Every component through F of the scheme of homogeneous maps $F' : \mathbb{A}^2(1, m) \rightarrow C$ satisfying $F'(0, 1) = P$ has dimension at least*

$$\chi = n + \frac{r}{m} - \sum_{j=1}^n \frac{b'_j}{m}. \quad (3.5)$$

In particular this dimension is at least n , except possibly when $r/m = N$ and the entire tangent action at P is trivial.

Proof. Homogeneous maps form an affine scheme of finite type: in fixed homogeneous generators of S , each coordinate of the map is a polynomial of a specified weighted degree, and the relations of S give polynomial equations in their finitely many coefficients. The condition of avoiding the vertex outside the origin is open, as may be checked on the proper weighted line. Near the given map, deformation theory can therefore be performed entirely in the smooth target C° .

Let E be the rank- n bundle on $\mathbb{P}(1, m)$ obtained by descending $F^*T_{C^\circ}$. Its determinant is $\mathcal{O}(r)$, so $\deg E = r/m$. We are deforming an equivariant map with its source and its grading torsor fixed; equivalently, we are deforming a section of the associated smooth bundle, with the canonical section obstruction theory of [49, Theorem 4.13 and Equation (4.18)]. There is no quotient by source automorphisms or by changes of the torsor identification. If D_∞ denotes the residual gerbe at infinity, the tangent and obstruction spaces for fixing its value are

$$H^0(\mathbb{P}(1, m), E(-D_\infty)), \quad H^1(\mathbb{P}(1, m), E(-D_\infty)).$$

Thus their Euler characteristic is a lower bound for the dimension of every component through the map.

For completeness, orbifold Riemann–Roch here says $\chi(E) = n + \deg E - \text{age}(E_\infty)$. It follows by taking the coarse pushforward: locally diagonalizing at the stack point subtracts the fractional character weights from the degree, after which ordinary Riemann–Roch on $\mathbb{P}^1$ applies. Restriction to D_∞ has Euler characteristic equal to the dimension of the invariant fiber, because μ_m is linearly reductive. Subtracting that invariant dimension is precisely the effect of representing each trivial character by m instead of zero. The exact sequence for restriction to D_∞ therefore gives Equation (3.5).

Since $\sum b'_j/m \leq n$, we have $\chi \geq r/m$. The Euler characteristic χ is an integer. If $r/m > N$, it is at least $N + 1 = n$. If $r/m = N$ and some character is nontrivial, the inequality is strict and the same conclusion follows. $\square$

3.2 Finite evaluation and its symmetric tensor

Assume that the exception in Lemma 3.2 does not occur. Let H be the reduced closure of an irreducible component of the open locus of maps avoiding the vertex outside the origin, in the affine scheme of homogeneous maps with $F'(0, 1) = P$. Then H is integral and

$$\dim H \geq n. \quad (3.6)$$

Precomposition with $(x, y) \mapsto (ax, y)$ preserves the chosen component because the acting group $\mathbb{G}_m$ is connected. It gives a positive grading on H : every coefficient involving x has positive weight, whereas the coefficients involving only y are fixed by P . All maps contract to the same map obtained by deleting the terms involving x . Consider the equivariant evaluation morphism

$$\mathrm{ev} : H \longrightarrow C, \quad F' \longmapsto F'(1, 0). \quad (3.7)$$

Lemma 3.3 (Finite evaluation). *The morphism in Equation (3.7) is finite and surjective. Its fiber over o is supported at the unique contracted map.*

Proof. Take $F_0 \in H$ with $F_0(1, 0) = o$. Choose a pointed curve in H through F_0 , with general point in the locus avoiding the vertex, and normalize it. Its generic map gives a twisted stable graph map to $B \times \mathbb{P}^1$: the second map is the coarse identification of $\mathbb{P}(1, m)$ with $\mathbb{P}^1$, and infinity is marked. This graph is stable even before any contraction because its second map has degree one. After a finite base change, properness of twisted stable maps gives a limiting graph map [1, Theorem 4.3]. The target is a tame proper smooth Deligne–Mumford stack with projective coarse space, as required by that theorem.

There is a unique component of the limiting coarse source of second degree one. It maps isomorphically to $\mathbb{P}^1$; call it the horizontal component. All other components form rational trees attached to it, since the total arithmetic genus is zero. The first map on the horizontal component is the coarse rational map defined by F_0 , after cancellation of its basepoints. Indeed the two maps agree on the open set where the limiting homogeneous coordinates do not vanish simultaneously.

We spell out why a positive-degree tail occurs over the finite basepoint $[1 : 0]$. Choose a sufficiently divisible integer k such that kL descends to a very ample bundle on the coarse space of B , and $m \mid k$. Its pullback along the original maps is the coarse bundle $\mathcal{O}_{\mathbb{P}^1}(k/m)$. The limiting projective coordinates are binary forms of that degree. They have a common zero at $[1 : 0]$, because $F_0(1, 0) = o$, and are not all zero, because $F_0(0, 1) = P \neq o$. Cancelling their common divisor loses positive degree at $[1 : 0]$. A general hyperplane avoiding the value of the canceled map at that point has the lost positive number of inverse-image points specializing there. In the limiting stable map these points cannot lie on the horizontal component. They are therefore accounted for by positive first degree on a tree over $[1 : 0]$. This also proves the degree assertion directly, without making an assumption about components that might occur over infinity.

Choose a terminal leaf of that tree. It does not contain the infinity marking, since its second image is $[1 : 0]$. It has one node and no markings. If its first map had degree zero, it would be constant in both factors and unstable: stability requires the induced coarse map to be a Kontsevich stable map [1, Section 4], and an unmarked constant terminal leaf has only one special point. Hence its first degree is positive. Its normalization is a twisted rational curve with at most one stack point: the only allowed stack points of a twisted source are nodes and markings. Projection from the product target to B preserves representability, since $\mathbb{P}^1$ has trivial inertia. The leaf consequently gives a representable nonconstant map $\mathbb{P}(1, \ell') \rightarrow B$ of the type used in the minimum.

Its L -degree is at least the original minimal degree $1/m$. Degree is additive on the limiting source and every component has nonnegative L -degree, while the total degree is $1/m$. Thus

this leaf consumes the full first degree. The horizontal component has first degree zero, and its constant value is the coarse image of P . It follows that the image of the affine map F_0 lies in the closure of the grading orbit of P .

The normalization of that orbit closure is $\mathbb{A}^1$, with a parameter of grading weight m , normalized to have value one at P . To see the weight, the gcd of the degrees of the coordinates nonzero at P is its stabilizer order m . The map from the normal source $\mathbb{A}^2$ factors through this normalization: its parameter is integral over the coordinate ring of the orbit curve and hence regular on $\mathbb{A}^2$. The parameter pulled back along F_0 is a weighted homogeneous polynomial of degree m , so it has the form

$$y + cx^m.$$

The coefficient of y is one because the value at $(0, 1)$ is fixed. Evaluation at $(1, 0)$ is the vertex only if $c = 0$. Therefore the vertex fiber is supported at the single contracted map.

Finally, let R_H be the positively graded coordinate ring of H . The preceding fiber statement says that R_H/S_+R_H is finite dimensional. Lift a homogeneous basis of this quotient. Induction on the positive degree shows that these lifts generate R_H as an S -module; this is graded Nakayama. Thus evaluation is finite. Its image is closed and has dimension $\dim H \geq n$, by Equation (3.6). Since C is integral of dimension n , the image is all of C , and $\dim H = n$. $\square$

Lemma 3.4 (A tensor of negative integral weight). *Let b be the degree of the finite evaluation in Lemma 3.3. There is a nonzero algebraic section*

$$s \in H^0(C^\circ, \text{Sym}^b T_{C^\circ})$$

of integral-grading weight $-bm$.

Proof. The source automorphisms $(x, y) \mapsto (x, y + tx^m)$, for $t \in \mathbb{C}$, preserve the homogeneous-map scheme, the fixed value at $(0, 1)$, and its open locus avoiding the vertex. The connected additive group therefore preserves the component H . Its vector field is nonzero: a coordinate nonzero at P has a nonzero pure power of y , whose derivative under this action is nonzero in characteristic zero. Differentiating evaluation gives a regular vector V along evaluation over C° . It is generically nonzero, since the finite dominant evaluation is generically separable and hence has injective differential on a nonempty smooth open set. For a target coordinate of degree d , its component is $\partial_y F'(1, 0)$, of coefficient-grading degree $d - m$. Thus V , as a vector along evaluation, has derivation weight $-m$.

Over the generic point, take the symmetric product of the vectors on all b sheets. This gives a nonzero element of $\text{Sym}^b T_{C^\circ}$ over $\mathbb{C}(C)$; the product, unlike the ordinary trace, cannot cancel. It is invariant under permutation of the sheets and so descends to that field. It is regular as well. On a smooth affine open set of C° trivializing the tangent bundle, the components of V lie in a finite algebra over the base ring. The coefficients of their symmetric product are therefore integral over the base ring and lie in its fraction field. Normality of the base makes them regular. This argument does not require H to be normal; alternatively one can first use its finite normalization. The construction is equivariant and multiplies weights by b , giving weight $-bm$. $\square$

3.3 A bound for homogeneous symmetric tensors

The tensor just constructed has negative weight for the integral grading. To exploit this, we first bound the Reeb weight of a homogeneous symmetric tensor. By the Fano cone Yau–Tian–Donaldson theorem [8, Theorem 1.1 and Definitions 2.1–2.2], C° carries a Ricci-flat Kähler cone metric with Reeb vector ξ . The theorem applies here because the puncture is smooth and the singularity is Q-Gorenstein and klt. The condition called K-stability in that reference allows equality only for an isomorphic central fiber, and is the K-polystability condition used here. With $A(\xi) = n$, the real homothetic dilations have infinitesimal generator $-J\xi$, and their pullback multiplies the metric by the square of the dilation factor.

Lemma 3.5 (Tensor weight bound). *Let $b \geq 1$, and let s be a nonzero homogeneous algebraic section of $\text{Sym}^b T_{C^\circ}$, of ξ -weight τ , with vector fields weighted as derivations. Then*

$$\tau \geq -b.$$

Proof. The Chern connection induced on $\text{Sym}^b T_{C^\circ}$ has zero mean curvature: contraction of the curvature on T_{C° is Ricci curvature, and contraction commutes with the induced tensor and symmetric power operations. Put $u = |s|^2$. The holomorphic Bochner formula and Cauchy–Schwarz give, with a consistent positive complex-Laplacian normalization,

$$\Delta u = |\nabla^{1,0} s|^2 \geq 0, \quad |\partial u|^2 \leq u \Delta u.$$

Consequently $\log(u + a)$ is subharmonic for each $a > 0$, and so is $(u + a)^{p/2}$ for every $p > 0$, by the chain rule applied to the increasing convex exponential function. Letting a decrease to zero shows that $|s|^p$ is subharmonic in the distributional sense, including across the zero set of s .

Let ρ be the metric radial coordinate and $M = \{\rho = 1\}$ its compact smooth link. A contravariant tensor of rank b and derivation weight τ has norm of radial degree $\tau + b$. If $\tau + b < 0$, choose $p > 0$ such that

$$d = p(\tau + b) \in (2 - 2n, 0).$$

Then $|s|^p = \rho^d \phi$ for a nonnegative continuous function ϕ on M with positive integral. The cone-Laplacian formula, valid also distributionally, is

$$\Delta(\rho^d \phi) = \rho^{d-2} (d(d + 2n - 2)\phi + \Delta_M \phi).$$

Integrating the angular term over M gives zero. Testing subharmonicity against a nonnegative radial function supported in an annulus would therefore imply $d(d + 2n - 2) \int_M \phi \geq 0$, which is impossible. Thus $\tau + b \geq 0$. $\square$

3.4 Uniform control of characters and the conclusion

The tensor rank in Lemma 3.4 can vary with the grading. The following estimate makes the approximation error independent of that rank.

Lemma 3.6 (Character estimate). *There is a constant $c > 0$ with the following property for all sufficiently close gradings in Equation (3.1). If a nonzero rank- b tensor on C° has integral-grading weight $-bm$, with $m > 0$, then it has a nonzero torus eigencomponent of character χ satisfying*

$$\|\chi\| \leq cb.$$

Consequently, if the tensor is as in Lemma 3.4, then

$$m \leq \frac{r}{n} + c\|\epsilon\|. \tag{3.8}$$

Proof. For the fixed eigen-generators f_j , positivity of ξ and absolute convergence of the error give constants $c_1, c_2 > 0$ such that their integral degrees $d_j = \alpha_j(\zeta)$ satisfy

$$c_1 r \leq d_j \leq c_2 r \quad \text{for every } j. \quad (3.9)$$

These bounds include the exact rational grading.

The differentials df_j span the cotangent bundle of C° . Pairing a symmetric tensor with all ordered b -tuples of these differentials embeds its space of sections into a finite tuple of regular functions on C° . Normality extends these functions over o . This also shows that the torus action on sections is locally finite, so the tensor has a nonzero torus eigencomponent; every such component retains its specified ζ -weight.

Let χ be the character of a nonzero component, and choose a nonzero pairing with $df_{j_1} \otimes \cdots \otimes df_{j_b}$. Its function has character $\beta = \chi + \alpha_{j_1} + \cdots + \alpha_{j_b}$ and integral degree

$$0 \leq d = -bm + d_{j_1} + \cdots + d_{j_b} \leq c_2 rb.$$

It has a polynomial lift in the generators consisting of monomials of character β , by torus equivariance of the polynomial presentation of S . Each such monomial contains at most $d/(c_1 r) \leq (c_2/c_1)b$ factors, by Equation (3.9). As the generator characters are fixed, this proves $\|\beta\| = O(b)$, and then $\|\chi\| = O(b)$, uniformly in the tensor, its rank and the grading.

Put $\tau = \chi(\xi)$. The eigencomponent satisfies Lemma 3.5, so $\tau \geq -b$. On the other hand,

$$-bm = \chi(\zeta) = \frac{r}{n}\tau + \chi(\epsilon) \geq -\frac{r}{n}b - cb\|\epsilon\|.$$

Dividing by b proves Equation (3.8). $\square$

Proof of Theorem 3.1. For each grading choose the minimal curve and its stabilizer order m as above. If the exceptional case in Lemma 3.2 occurs, then $r/m = N$ and the μ_m -action on $T_{C,P}$ is trivial. Faithfulness of the transverse tangent action implies $m = 1$, hence $r = N$. Equation (3.2) then forces $m_{\max} = 1$.

For the remaining indices, Lemmas 3.3 and 3.4 apply. The character estimate and Equation (3.4) give

$$0 \leq nm - r \leq nc\|\epsilon\|.$$

The left side is an integer. Since the error tends to zero, we have $nm = r$ for all sufficiently large indices. For a rational ray with its constant exact grading the error is zero, so this equality holds directly.

Now the finite evaluation has $\dim H = n$, whereas its deformation dimension satisfies

$$n = \dim H \geq n + \frac{r}{m} - \sum_{j=1}^n \frac{b'_j}{m} = 2n - \sum_{j=1}^n \frac{b'_j}{m} \geq n.$$

All equalities must hold. In particular every $b'_j = m$, so the tangent action is trivial. Its faithfulness again gives $m = 1$, and now $r = n = N + 1$. Finally

$$m_{\max} \leq \frac{N+1}{N} < 2$$

forces $m_{\max} = 1$.

The quotient is therefore an ordinary smooth projective Fano N -fold and L an actual ample line bundle. The Kobayashi–Ochiai theorem in its ample-line-bundle formulation [22], also supplied by [2, Theorems 1.1 and 1.3], applied to $-K_B = rL$ with $r = N + 1$ or N , gives precisely the two polarized varieties stated in the theorem. The polarization is the standard

one, as also follows by substituting into their Picard groups. By the section-ring identity from the cone setup,

$$S = \bigoplus_{j \geq 0} H^0(B, jL).$$

This is the polynomial ring for $(\mathbb{P}^N, \mathcal{O}(1))$, and the homogeneous coordinate ring of a nondegenerate quadratic hypersurface for $(Q^N, \mathcal{O}(1))$. Its affine cone is respectively smooth or the n -dimensional ordinary double point. $\square$

4 Jet cones and the canonical interior point

We work with the fixed cone supplied by Proposition 2.5. Thus $C = \text{Spec } S$ has dimension $n = N + 1$, its puncture is smooth, and its canonical module has a homogeneous nowhere-vanishing generator on the puncture. The minimizing Reeb vector ξ is normalized by $A_C(\xi) = n$. Throughout this section,

$$\widehat{\text{vol}}(o, C) \geq 2N^n. \quad (4.1)$$

Take the primitive positive gradings of Lemma 2.6, so that

$$\zeta_\nu = (r_\nu/n)\xi + \epsilon_\nu, \quad r_\nu = A_C(\zeta_\nu), \quad \epsilon_\nu \rightarrow 0$$

in absolute cocharacter norm. In the rational case we use the exact constant grading. Consider a subsequence on which the largest stabilizer order satisfies $m_\nu \geq r_\nu/N$, and choose a point attaining that order. We suppress the index ν when discussing a single grading. Constants carrying a subscript C may depend on this fixed cone and on N ; no uniformity over different cones is needed.

4.1 Leading Taylor terms and the real cone

Let B and L be the smooth quotient orbifold and its weight-one line bundle from Lemma 2.7. In particular,

$$S_j = H^0(B, jL), \quad -K_B = rL.$$

At the chosen point, take a smooth equivariant uniformizer with parameters $z_1, \dots, z_N$ and stabilizer μ_m . Choose the parameters and the frame of L so that their characters are respectively $b_1, \dots, b_N \in \mathbb{Z}/m$ and 1. We choose these algebraically: averaging regular lifts of a cotangent eigenbasis gives an equivariant map to $\mathbb{A}^N$ that is étale at the chosen point, and averaging a nonvanishing local frame gives the specified line character. After shrinking, these are equivariant algebraic étale coordinates and an algebraic frame. A section of jL is then represented by a power series $F(z)$ satisfying

$$F(\varepsilon^{b_1} z_1, \dots, \varepsilon^{b_N} z_N) = \varepsilon^j F(z).$$

Order monomials first by total degree and then lexicographically, with the smallest monomial first. Write $\text{in}(s) = z^\gamma$ for the leading monomial, omitting its nonzero scalar coefficient, and put

$$\Gamma = \{(j, \gamma) : 0 \neq s \in S_j, \text{in}(s) = z^\gamma\}, \quad (4.2)$$

$$\Lambda = \{(j, \gamma) \in \mathbb{Z} \times \mathbb{Z}^N : j \equiv \sum_i b_i \gamma_i \pmod{m}\}, \quad (4.3)$$

$$K = \overline{\text{cone}(\Gamma)} \subset \mathbb{R} \times \mathbb{R}^N. \quad (4.4)$$

Here $(0, 0) \in \Gamma$, and $\gamma = 0$ denotes a nonzero constant Taylor term, including those of sections of positive degree. The degree coordinate in Equations (4.2)–(4.4) is the *original* degree j .

For integrations only, introduce the scaled degree and the number

$$s = \frac{r}{Nm} \leq 1, \quad D_0 = \frac{j}{sm} = \frac{Nj}{r}. \quad (4.5)$$

We distinguish the coordinate D_0 from j throughout. Write $\widetilde{K}$, $\widetilde{\Gamma}$, and $\widetilde{\Lambda}$ for the images of K , Γ , and Λ under $(j, x) \mapsto (Nj/r, x)$.

Proposition 4.1 (Jet setup). *The following assertions hold.*

- (i) Γ is an additive semigroup with one-dimensional leaves, and its generated group is exactly Λ . Every finite compatible Taylor jet is realized by sections of all sufficiently large degrees in each admissible progression.
- (ii) K is a full-dimensional closed convex cone, contains the positive constant ray, projects onto the whole exponent orthant, and satisfies $K = \text{int } \widetilde{K}$. Its bounded-degree sections are bounded. More precisely, uniformly in the chosen point and nearby grading,

$$|\gamma| := \sum_i \gamma_i \leq C_C \frac{j}{r} \quad ((j, \gamma) \in K). \quad (4.6)$$

Its scaled image has the form

$$\widetilde{K} = \{(D_0, x) : x \in \mathbb{R}_{\geq 0}^N, D_0 \geq f(x)\}, \quad (4.7)$$

where f is finite, positively homogeneous, convex, and

$$f(x) \geq c_C |x| \quad (x \in \mathbb{R}_{\geq 0}^N) \quad (4.8)$$

for a constant $c_C > 0$ independent of the nearby grading and chosen point.

- (iii) In the scaled coordinates the lattice density is s . If ℓ is a linear functional strictly positive on $\widetilde{K} \setminus \{0\}$, then

$$\begin{aligned} & \#\{\gamma' \in \widetilde{\Gamma} : \ell(\gamma') < q\} \\ &= s \text{Leb}\{(D_0, x) \in \widetilde{K} : \ell(D_0, x) < 1\} q^n + o(q^n), \end{aligned} \quad (4.9)$$

with Lebesgue measure in the scaled coordinates. No saturation assumption on Γ is required.

- (iv) The vector

$$c_0 = (r, \mathbf{1}) \quad \text{in the original } (j, \gamma) \text{ coordinates} \quad (4.10)$$

belongs to Λ . Its scaled coordinates are $(N, \mathbf{1})$.

Proof. Leading monomials multiply, and the associated leading coefficient is a scalar. This proves additivity and the one-dimensional-leaf assertion. In particular, each leading index in a fixed degree accounts for one dimension of S_j .

Choose an integer $M > 0$ divisible by all stabilizer orders such that ML descends to an ample line bundle on the coarse space. In a fixed residue class j_0 modulo M , prescribe a finite compatible Taylor jet of a section of $j_0 L$. The kernel of the map to that finite jet space is coherent. Serre vanishing after tensoring by sufficiently large powers of ML , or after exact tame pushdown to the coarse space, makes the map on global sections surjective. Thus the prescribed jet is realized in every sufficiently large degree $j_0 + kM$. Taking all the

finitely many admissible residue classes proves the assertion about progressions. In particular, constants occur in all sufficiently large degrees divisible by m , and each monomial z^γ occurs as a leading term in sufficiently large compatible degrees.

Differences of two such constant indices whose degrees differ by m give $(m, 0)$ in the generated group. For each i , a realized linear jet gives an index (j_i, e_i) with $j_i \equiv b_i \pmod{m}$. The vectors $(m, 0), (j_1, e_1), \dots, (j_N, e_N)$ generate the full congruence lattice in Equation (4.3). All indices satisfy that congruence, proving equality of groups. These same indices show that the cone contains the positive constant ray, has full dimension, and projects onto the exponent orthant.

We next prove the uniform estimate. Fix torus eigen-generators of S and the resulting affine embedding of C . Their ζ_ν -degrees are bounded below by a fixed positive multiple of r_ν for all sufficiently large ν , because their ξ -weights are positive. A section of degree j has a homogeneous polynomial lift in these generators, of ordinary polynomial degree at most Cj/r . At a smooth point of the punctured cone, a homogeneous function has the same local vanishing order as its restriction to a transverse slice: in orbit and slice coordinates it is a unit power of the orbit coordinate times its slice expression.

Take a general complete-intersection curve through the smooth point, cut out by affine hyperplanes in the fixed embedding. It is smooth there and is not contained in the zero locus of the nonzero function. The local intersection multiplicity with the function is at least its local vanishing order. Bézout's inequality bounds this multiplicity by the polynomial degree times the degree of the fixed embedded cone. This proves Equation (4.6) for Γ , and then for K by homogeneity and closure. In particular, bounded degree cuts are compact.

The positive constant ray makes every nonempty fiber over an exponent x an upper ray in degree. Every exponent fiber is nonempty, since nonnegative combinations of the indices (j_i, e_i) reach every $x \geq 0$. Closedness and the degree bound make the lowest degree in each fiber attainable. Hence K is the epigraph of a finite convex positively homogeneous function in the scaled coordinates. Equation (4.6) becomes Equation (4.8). A full-dimensional closed convex cone is the closure of its interior.

The original lattice has covolume m . The coordinate change $(j, \gamma) \mapsto (Nj/r, \gamma)$ multiplies covolumes by N/r ; the resulting covolume is $Nm/r = 1/s$. For clarity, we give the nonsaturated semigroup counting argument. This is the conductor-and-exhaustion method for value semigroups [19, Theorems 1.4 and 1.6], also used for graded linear series in [25, Proposition 2.1]. Its asymptotic conclusion does not assert exact saturation. Let Γ_0 be a finitely generated subsemigroup with full generated group Λ , and let K_0 be its cone. There is a vector $c \in \Gamma_0$ such that

$$c + (K_0 \cap \Lambda) \subset \Gamma_0. \quad (4.11)$$

Indeed every lattice point of K_0 lies in a cone generated by a linearly independent subset of the finite generating set. Removing the integer parts of its nonnegative coefficients writes it as a semigroup element plus a lattice point in one of finitely many bounded parallelepipeds. There are only finitely many possible remainders v . Since the group is Λ , write each such remainder as $v = v_+ - v_-$ with $v_\pm \in \Gamma_0$. The sum of all the v_- is a single c for which Equation (4.11) holds. Translating a bounded linear cutoff changes its scale by a constant, so ordinary lattice counting between $c + (K_0 \cap \Lambda)$ and $K_0 \cap \Lambda$ gives its leading lattice volume. After the coordinate change this is Euclidean volume with density s .

Now exhaust $\text{int } K$ by cones of finitely many indices of Γ , always retaining a fixed finite set whose group is Λ . Every interior point belongs to the interior of one of these finite cones: choose a small simplex surrounding it in $\text{int } K$ and approximate the simplex vertices by elements of $\text{cone}(\Gamma)$. The resulting finite cones give the lower leading count. Counting all lattice points of K gives the upper count. Compact convex cutoff bodies have boundary of Lebesgue measure zero, and their volumes are exhausted from the interior. This proves

Equation (4.9).

Finally, on the uniformizer the canonical character is the negative of the sum of the coordinate characters, in the convention for equivariant line bundles used here. The identity $-K_B = rL$ therefore gives $r \equiv \sum_i b_i \pmod{m}$, proving Equation (4.10). $\square$

4.2 Discrepancy and moment tests

Lemma 4.2 (The degree and Taylor-order test). *For $\delta \in \mathbb{Q}_{\geq 0}$, the rule*

$$v_\delta\left(\sum_j s_j\right) = \min_{s_j \neq 0} \left\{ \frac{Nj}{r} + \delta \operatorname{ord}_z(s_j) \right\} \quad (4.12)$$

defines a valuation centered at the vertex. It is divisorial up to positive rescaling, and

$$A_C(v_\delta) = N + N\delta, \quad (4.13)$$

$$A_C(v_\delta)^n \operatorname{vol}(v_\delta) = N^n (1 + \delta)^n s \int_{\mathbb{R}_{\geq 0}^N} e^{-f(x) - \delta|x|} dx. \quad (4.14)$$

Proof. Use the local line-bundle extraction over the chosen uniformizer. Write t for its line coordinate, of grading weight one. Its finite quotient presentation has the action

$$(t, z_1, \dots, z_N) \mapsto (\varepsilon^{-1}t, \varepsilon^{b_1}z_1, \dots, \varepsilon^{b_N}z_N).$$

A homogeneous function of degree j pulls back to $t^j F_j(z)$. The monomial valuation with costs

$$v(t) = N/r = 1/(sm), \quad v(z_i) = \delta$$

therefore restricts to Equation (4.12). Distinct original degrees have different powers of t , so even when scalar values agree their lowest monomials cannot cancel. Rational positive costs give a divisorial monomial valuation up to rescaling. At $\delta = 0$ the rule is the rescaled ordinary degree valuation. The positive degree cost and the positivity of all algebra-generator degrees imply that its center is the vertex.

Let Ω_C be the homogeneous canonical generator of weight r . On this chart it has the form

$$t^{r-1}u(z) dt \wedge dz_1 \wedge \dots \wedge dz_N, \quad u(0) \neq 0.$$

The exponent follows from homogeneity, and the coefficient is a unit because Ω_C is nonvanishing on the punctured cone. Computing discrepancy from this pulled-back form, including the finite-chart ramification in the restricted value scaling, gives

$$A_C(v_\delta) = \frac{N}{r} + N\delta + (r-1)\frac{N}{r} = N + N\delta.$$

Equivalently one can take a weighted monomial extraction on the smooth chart and apply the canonical divisor formula there; the same calculation descends to the quotient. For $\delta = 0$ it is the canonical degree formula.

In each S_j , a leading-term basis is compatible with total Taylor order. Thus the colength below a cutoff q is the number of indices of Γ satisfying $D_0 + \delta|x| < q$. These finite quotients are supported at the vertex, so computing them in S or in its local ring gives the same length. Proposition 4.1 applies to this strictly positive cutoff functional. Since K has dimension n , homogeneity and integration by layers give

$$n! \operatorname{Leb}\{(D_0, x) \in \widetilde{K} : D_0 + \delta|x| < 1\} = \int_{\widetilde{K}} e^{-D_0 - \delta|x|} dD_0 dx = \int_{\mathbb{R}_{\geq 0}^N} e^{-f(x) - \delta|x|} dx.$$

Combining this identity with the lattice density and Equation (4.13) proves Equation (4.14). $\square$

Let $X_1, \dots, X_N$ be independent exponential random variables of mean one. For the current grading put

$$B_0(x) = f(x) - |x|, \quad H = \mathbb{E}e^{-B_0(X)} = \int_{\mathbb{R}_{\geq 0}^N} e^{-f(x)} dx. \quad (4.15)$$

Lemma 4.3 (The two moment inequalities). *Along the fixed-cone sequence of gradings, there are numbers $\eta_\nu \geq 0$ tending to zero such that*

$$H \geq \frac{2}{s}, \quad \mathbb{E}[(1 + B_0(X))e^{-B_0(X)}] \geq -\eta_\nu H. \quad (4.16)$$

For an exact minimizing rational grading one may take $\eta_\nu = 0$.

Proof. Write $F(\delta)$ for the right side of Equation (4.14). The volume hypothesis and the test at zero give

$$2N^n \leq \widehat{\text{vol}}(o, C) \leq F(0) = N^n s H,$$

which is the first inequality.

Consider the probability measure $d\mu_f = H^{-1}e^{-f(x)}dx$. Positive homogeneity gives, for every $t > 0$,

$$\int e^{-tf(x)} dx = t^{-N} H.$$

Differentiating, justified by Equation (4.8), yields

$$\mathbb{E}_{\mu_f} f = N, \quad \mathbb{E}_{\mu_f} f^2 = N(N+1). \quad (4.17)$$

In particular,

$$\begin{aligned} \frac{F'(0)}{F(0)} &= n - \mathbb{E}_{\mu_f} |x| \\ &= 1 + \mathbb{E}_{\mu_f} (f - |x|) = \frac{\mathbb{E}[(1 + B_0)e^{-B_0}]}{H}. \end{aligned} \quad (4.18)$$

We check the uniformity needed for near-minimality. Equation (4.8) and Equation (4.17) give uniform first and second moments of $|x|$ under μ_f . Indeed they are at most $c_C^{-1}N$ and $c_C^{-2}N(N+1)$, respectively. Since

$$\frac{F(\delta)}{F(0)} = (1 + \delta)^n \mathbb{E}_{\mu_f} e^{-\delta|x|},$$

twice differentiating shows

$$\sup_{0 \leq \delta \leq 1} \frac{|F''(\delta)|}{F(0)} \leq C'_C \quad (4.19)$$

for a fixed finite constant. The three terms are bounded by constant multiples of 1, $\mathbb{E}_{\mu_f} |x|$, and $\mathbb{E}_{\mu_f} |x|^2$, because $e^{-\delta|x|} \leq 1$.

The test at zero is a rescaling of the current degree valuation, so Lemma 2.6 gives $F_\nu(0) \rightarrow \widehat{\text{vol}}(o, C)$. If $F'_\nu(0)/F_\nu(0)$ were bounded above by a fixed negative number along a subsequence, Taylor's formula and Equation (4.19) would give a fixed small positive rational δ and a fixed $\rho > 0$ such that $F_\nu(\delta) \leq (1 - \rho)F_\nu(0)$. This is incompatible with $F_\nu(\delta) \geq \widehat{\text{vol}}(o, C) > 0$ and the preceding convergence. Thus $F'_\nu(0)/F_\nu(0) \geq -\eta_\nu$ for some $\eta_\nu \rightarrow 0$. Equation (4.18) proves the second inequality. At an exactly minimizing degree valuation, every positive rational test has volume at least $F(0)$, so its right derivative is nonnegative. $\square$

4.3 A uniform exponential expectation inequality

We need to bound H when B_0 is bounded below by a linear function whose coefficients sum to zero. The second inequality in Lemma 4.3 controls the expectation of $(1 + B_0)e^{-B_0}$. Combining it with H leads, for $u > 0$, to $(1 + (1 + B_0)/u)e^{-B_0}$, whose pointwise upper envelope is the truncated exponential below.

Define the continuously differentiable function

$$g(y) = \begin{cases} 1, & y \leq 0, \\ (1 + y)e^{-y}, & y > 0, \end{cases} \quad (4.20)$$

Here $0 \leq g \leq 1$.

For any real v and any $u > 0$, elementary differentiation gives

$$\sup_{b \geq v} \left(1 + \frac{1+b}{u}\right) e^{-b} = \frac{e^u}{u} g(u+v). \quad (4.21)$$

Indeed the derivative of the expression on the left before maximization is $-(u+b)e^{-b}/u$, so its maximum is attained at $b = -u$ when $v \leq -u$, and at $b = v$ otherwise.

Theorem 4.4 (The exponential bound). *For $u = 4/3$ one has*

$$\sup_{N \geq 1} \sup_{\substack{a \in \mathbb{R}^N \\ \sum_i a_i = 0}} \frac{e^u}{u} \mathbb{E} g(u + a \cdot X) = C_* := \frac{3}{4} e^{4/3} - \frac{e}{3} < 2. \quad (4.22)$$

The supremum over dimensions is approached by the vectors $(2, -2/(N-1), \dots, -2/(N-1))$ as $N \rightarrow \infty$.

The theorem gives a bound strictly below the threshold 2 in Lemma 4.3. We prove it before applying it to the canonical point. For the proof, put $h = 1 - g$; then

$$h'(y) = ye^{-y} \mathbf{1}_{\{y > 0\}}, \quad k(y) := h''(y) = (1 - y)e^{-y} \mathbf{1}_{\{y > 0\}} \quad (4.23)$$

holds almost everywhere. In particular h' is globally Lipschitz and k is bounded.

Lemma 4.5 (Log-concavity and a barycenter half-space bound). *A nonzero mean-zero linear combination of independent mean-one exponentials has a positive continuous log-concave density on the whole real line. Adding further independent scaled exponentials, with either sign of coefficient, preserves these properties. If Z has such a density, then*

$$\frac{1}{e} \leq \mathbb{P}(Z \leq \mathbb{E}Z) \leq 1 - \frac{1}{e}. \quad (4.24)$$

Proof. The preservation of log-concavity under convolution is a special case of the classical theory of Prékopa [45, Theorems 7–8]; we give the elementary exponential-specific argument needed here. A nonzero coefficient vector with sum zero has a positive and a negative coefficient. The difference $aE - bE'$ for $a, b > 0$ has density

$$p(x) = \frac{1}{a+b} \begin{cases} e^{x/b}, & x \leq 0, \\ e^{-x/a}, & x \geq 0. \end{cases}$$

It is positive, continuous, and log-concave.

Let r be a positive log-concave function for which the indicated one-sided integrals are finite. Set $R(x) = \int_{-\infty}^x r(y)dy$. For $t \geq 0$, concavity of $\log r$ implies that $r(x-t)/r(x)$ is nondecreasing in x . Thus

$$\frac{R(x)}{r(x)} = \int_0^\infty \frac{r(x-t)}{r(x)} dt$$

is nondecreasing, and $(\log R)' = r/R$ is nonincreasing. Consequently R is log-concave. The same argument with $r(x+t)/r(x)$ shows that $T(x) = \int_x^\infty r(y)dy$ is log-concave: T/r is nonincreasing, so $(\log T)' = -r/T$ is nonincreasing. These statements can equivalently be read almost everywhere; the positive functions involved are locally absolutely continuous.

If p is a positive log-concave probability density, adding cE or $-cE$, $c > 0$, produces respectively the densities

$$\frac{e^{-x/c}}{c} \int_{-\infty}^x e^{y/c} p(y) dy, \quad \frac{e^{x/c}}{c} \int_x^\infty e^{-y/c} p(y) dy.$$

The integrands are exponential tilts of p , hence log-concave, and the appropriate one-sided integrals are finite. The preceding argument proves log-concavity of the new densities. Positivity and continuity are immediate from the displayed formulas. Induction starting with $aE - bE'$ proves the assertions about finite exponential sums, and translation does not affect them.

For completeness, Equation (4.24) is the one-dimensional log-concave form of Grünbaum's centroid inequality [15]; see [37, Lemma 5.4] and [40, Section 3]. It follows directly here from the same integral argument. If G is the distribution function of Z , both G and $1 - G$ are log-concave. Since $G(Z)$ is uniform on $(0, 1)$, Jensen's inequality gives

$$\log G(\mathbb{E}Z) \geq \mathbb{E} \log G(Z) = \int_0^1 \log t \, dt = -1.$$

Applying the same reasoning to $1 - G$ gives $\log(1 - G(\mathbb{E}Z)) \geq -1$. All the means in the present application are finite, and both logarithmic expectations just used equal -1 . These two inequalities give Equation (4.24). $\square$

Lemma 4.6 (Strict single crossing). *Let Z have a positive log-concave probability density on $\mathbb{R}$, and let E be an independent mean-one exponential. If*

$$\mathbb{E}k(Z + c_*E) = 0,$$

then $\mathbb{E}k(Z + cE) > 0$ for every $c < c_$ and $\mathbb{E}k(Z + cE) < 0$ for every $c > c_*$.*

Proof. Write p for the density of Z and first consider

$$\psi(t) = \mathbb{E}k(Z + t) = \int_0^\infty (1 - y)e^{-y} p(y - t) dy.$$

This is continuous, by continuity of translation in L^1 and boundedness of k . A positive log-concave density has a continuous representative whose logarithm is finite and concave. For $t_1 < t_2$, the ratio

$$r(y) = p(y - t_2)/p(y - t_1)$$

is nondecreasing in y . If $\psi(t_1) = 0$, subtraction of the constant $r(1)$ gives

$$\psi(t_2) = \int_0^\infty k(y)p(y - t_1)(r(y) - r(1))dy \leq 0 :$$

the integrand is nonpositive on both $(0, 1)$ and $(1, \infty)$. Equality would force r to be constant on $(0, \infty)$, by positivity and continuity. Thus $\log p$ would have constant increments across

the fixed interval length $t_2 - t_1$ on a half-line. Concavity forces it to be affine there: its almost-everywhere derivative is nonincreasing, and equality of this derivative across a fixed shift makes it constant on that half-line. Integrability then gives $p(y - t_1) = Ae^{-\lambda y}$ for $y > 0$, with $A, \lambda > 0$. But

$$\psi(t_1) = A \int_0^\infty (1 - y)e^{-(1+\lambda)y} dy = \frac{A\lambda}{(1+\lambda)^2} > 0,$$

a contradiction. Hence the inequality is strict. Reversing the likelihood-ratio comparison proves strict positivity to the left of a zero. Continuity shows that ψ either has one strict sign everywhere or has exactly one zero, with positive sign to its left and negative sign to its right.

Now put $F(c) = \mathbb{E}\psi(cE)$. For $c \neq 0$, the density of cE is

$$q_c(t) = |c|^{-1}e^{-t/c}\mathbf{1}_{\{tc>0\}}.$$

If $c_1 < c_2$ have the same sign, then on their common support

$$\frac{q_{c_2}(t)}{q_{c_1}(t)} = \frac{|c_1|}{|c_2|} \exp(t(1/c_1 - 1/c_2))$$

is strictly increasing in t . This holds on the negative half-line as well as on the positive half-line. If $F(c_*) = 0$ and $c_* \neq 0$, then ψ must have its strict crossing inside that half-line. Subtracting the likelihood ratio at this crossing proves strict positivity of $F(c)$ for $c < c_*$ and strict negativity for $c > c_*$ within the same parameter half-line.

The comparisons across zero are also strict. If $c_* > 0$, the crossing of ψ lies in $(0, \infty)$, so $\psi > 0$ on $(-\infty, 0]$ and $F(c) > 0$ for $c \leq 0$. If $c_* < 0$, the crossing lies in $(-\infty, 0)$, giving $F(c) < 0$ for $c \geq 0$. Finally $F(0) = \psi(0)$; a zero there gives the two required signs directly from those of ψ . This proves the assertion for all real parameters. $\square$

Lemma 4.7 (Finite critical points). *Fix $u > 0$ and $N \geq 2$. Every nonzero finite local maximum of*

$$a \mapsto \mathbb{E}g\left(u + \sum_{i=1}^N a_i X_i\right), \quad \sum_i a_i = 0,$$

has, after permutation, the form

$$a = (t, -t/(N-1), \dots, -t/(N-1)) \quad (t > 0). \quad (4.25)$$

Proof. Equivalently consider a local minimum of $\mathbb{E}h(Y)$, where $Y = u + \sum_i a_i X_i$. At a nonzero coefficient vector, Lemma 4.5 supplies a continuous positive log-concave density for Y . First and second differentiation under the expectation are justified by boundedness of h' , its global Lipschitz property, the finite second moments of the variables X_i , and the absence of an atom at $Y = 0$. Stationarity in the direction $e_i - e_j$ gives

$$0 = \mathbb{E}[(X_i - X_j)h'(Y)] = (a_i - a_j)\mathbb{E}[X_i X_j k(Y)]. \quad (4.26)$$

To see the second equality, condition on all other variables and on $S = X_i + X_j$. Given $S = s$, X_i is uniform on $[0, s]$ and $X_j = s - X_i$. Integration by parts with $x(s - x)$, whose endpoint values vanish and whose derivative is $s - 2x$, gives exactly Equation (4.26).

For distinct indices, weighting by $X_i X_j$ changes those two independent exponential laws into independent Gamma(2, 1) laws. Since $\mathbb{E}X_i X_j = 1$, no normalization factor appears. Thus

$$\mathbb{E}[X_i X_j k(Y)] = \mathbb{E}k(Y + a_i E' + a_j E''), \quad (4.27)$$

where the added exponentials are independent of all variables in Y . If three distinct coefficient values occurred, fix an index i at one value and choose indices at the other two. Equations (4.26) and (4.27) would give two distinct zeros of

$$c \mapsto \mathbb{E}k(Y + a_i E' + c E'').$$

Its base variable has a positive full-support log-concave density by Lemma 4.5, contradicting Lemma 4.6. Therefore there are at most two coefficient values.

Suppose the two values are $b < c$ and the higher value occurs at indices $i \neq j$. A high-low pair gives a zero at b of $d \mapsto \mathbb{E}k(Y + c E' + d E'')$. Strict single crossing makes its value at c negative. By Equation (4.27), this says $\mathbb{E}[X_i X_j k(Y)] < 0$. As $a_i = a_j$, Y depends on this pair only through S , and

$$\mathbb{E}[(X_i - X_j)^2 \mid S] = S^2/3, \quad \mathbb{E}[X_i X_j \mid S] = S^2/6.$$

Consequently the Hessian in the permissible splitting direction is

$$\mathbb{E}[(X_i - X_j)^2 k(Y)] = 2\mathbb{E}[X_i X_j k(Y)] < 0,$$

contradicting a minimum. The higher value therefore occurs once. The sum-zero constraint forces it to be positive and gives Equation (4.25). $\square$

The preceding lemmas leave only one possible shape for a finite nonzero optimizer. To obtain a bound uniform in the dimension, it remains to control escape to infinity and then solve the resulting one-variable maximization.

Proof of Theorem 4.4. Put $\Phi_N(a) = (e^u/u)\mathbb{E}g(u + a \cdot X)$ and let $\mathcal{P}_N$ be its supremum under the sum-zero constraint. First control escape to infinity with N fixed. From an unbounded sequence a_k , pass to a subsequence with $a_k/\|a_k\| \rightarrow b \neq 0$. The vector b has sum zero, so $b \cdot X$ has mean zero and a continuous log-concave density. Almost surely away from its zero set,

$$g(u + a_k \cdot X) \rightarrow \mathbf{1}_{\{b \cdot X < 0\}}.$$

Bounded convergence and Equation (4.24) imply that every such subsequential limit is at most

$$B_* := \frac{e^u}{u}(1 - 1/e). \tag{4.28}$$

Thus, if $\mathcal{P}_N > B_*$, a maximizing sequence is bounded and a finite maximizer exists by continuity. The zero vector has value $1 + 1/u = 7/4$. Every other finite maximizer has the form in Lemma 4.7. If $\mathcal{P}_N \leq B_*$, the boundary estimate already suffices; we make no assertion of attainment in that case.

Appending a zero coefficient shows that $\mathcal{P}_N$ is nondecreasing in N , and it is bounded by e^u/u . Consider its limit. If that limit is at most $\max(B_*, 7/4)$, those two values suffice. Otherwise sufficiently large dimensions have finite nonzero maximizers. After permutation write them as in Equation (4.25), with exceptional coefficient $t_N > 0$. On a common probability space their arguments are

$$Y_N = u + t_N \left(X_1 - \frac{1}{N-1} \sum_{i=2}^N X_i \right).$$

Pass to a subsequence on which t_N converges in $[0, \infty]$. If $t_N \rightarrow t < \infty$, the law of large numbers and bounded convergence give

$$\Phi_N(a_N) \rightarrow \Phi_\infty(t) := \frac{e^u}{u} \mathbb{E}g(u + t(X_1 - 1)).$$

If $t_N \rightarrow \infty$, then $Y_N/t_N \rightarrow X_1 - 1$ almost surely. The limit has no atom at zero; on its positive and negative sets, respectively, Y_N tends to $+\infty$ and $-\infty$. Hence bounded convergence gives

$$\Phi_N(a_N) \longrightarrow \frac{e^u}{u} \mathbb{P}(X_1 < 1) = B_*.$$

This argument uses no comparison between the growth of t_N and $\sqrt{N}$.

It remains to maximize the scalar expression. For $0 \leq t \leq u$, the argument $u + t(X_1 - 1)$ is nonnegative, and direct integration gives

$$\Phi_\infty(t) = \begin{cases} \frac{e^t}{u} \left(\frac{1+u-t}{1+t} + \frac{t}{(1+t)^2} \right), & 0 \leq t \leq u, \\ \frac{e^u}{u} \left(1 - e^{-1+u/t} \frac{t^2}{(1+t)^2} \right), & t \geq u. \end{cases} \quad (4.29)$$

For the second branch, split the exponential integral at $X_1 = 1 - u/t$; above that point translate the integration variable so that the argument of g is t times the new variable. The integral there is $e^{-1+u/t}((1+t)^{-1} + t(1+t)^{-2})$, which yields the displayed formula.

The derivative of the first branch has the sign of $t(-t^2 + ut + u - 1)$. For $u = 4/3$, the second factor is positive throughout $[0, u]$, as its endpoint values are positive and it is concave. On the second branch, the logarithmic derivative of the positive quantity being subtracted is

$$-\frac{u}{t^2} + \frac{2}{t} - \frac{2}{1+t} = \frac{(2-u)t - u}{t^2(1+t)}.$$

Thus this branch is maximal at $t = u/(2-u) = 2$, and its value is

$$\Phi_\infty(2) = \frac{3}{4}e^{4/3} - \frac{e}{3} = C_*.$$

In particular $C_* > \Phi_\infty(0) = 7/4$. Moreover

$$C_* - B_* = \frac{3}{4}e^{1/3} - \frac{e}{3} > 0,$$

because $e^{1/3} > 1 + 1/3$ and $e < 3$. The preceding compactness and limit argument proves the upper bound $\sup_N \mathcal{P}_N \leq C_*$. Taking $t_N = 2$ and applying the law of large numbers proves the reverse inequality and the asserted approach to the supremum.

Finally the strict gap has an elementary rational certificate. The exponential series gives $e > 1359/500$ and $e^{4/3} < 1897/500$. For the latter bound one may retain terms through degree eight and bound the remaining tail by its first term divided by $1 - (4/3)/10$. Therefore

$$C_* < \frac{3}{4} \frac{1897}{500} - \frac{1}{3} \frac{1359}{500} = \frac{3879}{2000} < 2.$$

□

4.4 Interiority of the canonical vector

Theorem 4.8 (Canonical interior point). *For every sufficiently large grading in the subsequence under consideration,*

$$c_0 = (r, \mathbf{1}) \in \text{int } K \quad \text{in the original degree coordinates.} \quad (4.30)$$

Equivalently, $f(\mathbf{1}) < N$. For an exact minimizing rational grading with $m \geq r/N$, the conclusion holds without passage to a limit.

Proof. Suppose that $f(\mathbf{1}) \geq N$. At the interior point $\mathbf{1}$ of the exponent orthant, choose a subgradient p of f . Positive homogeneity gives $p \cdot \mathbf{1} = f(\mathbf{1})$ and $f(x) \geq p \cdot x$ on the orthant. Therefore $B_0(x) \geq (p - \mathbf{1}) \cdot x$, and the coefficient sum of $p - \mathbf{1}$ is nonnegative. Subtracting the same nonnegative number from each coefficient gives a vector a with sum zero and

$$B_0(x) \geq a \cdot x \quad (x \in \mathbb{R}_{\geq 0}^N). \quad (4.31)$$

Take $u = 4/3$. Applying Equation (4.21) pointwise to Equation (4.31), and using Lemma 4.3, gives

$$\begin{aligned} (1 - \eta_\nu/u)H &\leq \mathbb{E} \left[\left(1 + \frac{1 + B_0(X)}{u} \right) e^{-B_0(X)} \right] \\ &\leq \frac{e^u}{u} \mathbb{E} g(u + a \cdot X) \leq C_* < 2. \end{aligned}$$

All expectations are finite by Equation (4.8). But $H \geq 2/s \geq 2$, so the displayed inequality is impossible once η_ν is sufficiently small. Hence $f(\mathbf{1}) < N$ for all sufficiently large gradings. At an exact minimizing rational grading, $\eta_\nu = 0$ gives the contradiction immediately.

The scaled coordinates of c_0 are $(N, \mathbf{1})$, and $\mathbf{1}$ is interior to the exponent orthant. The strict epigraph inequality is therefore precisely Equation (4.30). Its membership in the lattice was already proved in Proposition 4.1. $\square$

5 Adjoint jets and the saturated semigroup

We retain the orbifold and jet data of Proposition 4.1. In particular, B is a smooth effective proper complex orbifold of dimension N , its coarse space is projective, L is an ample orbifold line bundle, and $-K_B = rL$. At the chosen point the stabilizer is μ_m , the fiber character of L is faithful, and equivariant transverse parameters $z_1, \dots, z_N$ have characters $b_1, \dots, b_N$. We use the original, unscaled degree coordinate:

$$\Lambda = \left\{ (j, \gamma) \in \mathbb{Z} \times \mathbb{Z}^N : j \equiv \sum_i b_i \gamma_i \pmod{m} \right\}, \quad c_0 = (r, \mathbf{1}).$$

The semigroup Γ records the total-degree-then-lexicographic Taylor initials of sections of jL , and K is its closed real cone. It has full dimension, contains the positive constant ray, and has bounded slices of bounded original degree. Its projection onto the exponent coordinates is the nonnegative orthant. The canonical character identity gives $c_0 \in \Lambda$.

The adjoint lifting statement below uses these data but does *not* assume that c_0 itself lies in the interior of K . We use Theorem 4.8 only after proving that statement.

5.1 Vanishing on the smooth orbifold models

We first record the precise form of vanishing needed on a weighted blowup and its resolution. Divisors and line bundles on a stack are written additively. Nefness and bigness of their rational classes are understood on the projective coarse space.

Lemma 5.1 (Orbifold vanishing). *Let $\mathcal{X}$ be a smooth proper Deligne–Mumford stack over $\mathbb{C}$, with trivial generic stabilizer and projective coarse space. Let Θ be an effective rational divisor with simple normal crossing support and coefficients less than one. If D is a Cartier divisor and $D - (K_{\mathcal{X}} + \Theta)$ is nef and big, then*

$$H^q(\mathcal{X}, \mathcal{O}_{\mathcal{X}}(D)) = 0 \quad (q > 0).$$

Proof. Let $q : \mathcal{X} \rightarrow X$ be the coarse morphism. For a prime divisor $B \subset X$, let e_B be the stabilizer order at the generic point of its inverse image. A rational divisor A on $\mathcal{X}$ has a coarse rational divisor A_c , characterized in codimension one by $q^*A_c = A$. Locally the coarse map is given in the normal direction by $t = z^{e_B}$, so

$$K_{\mathcal{X}} = q^*(K_X + R), \quad R = \sum_B (1 - 1/e_B)B.$$

Only divisors with $e_B > 1$ contribute to R .

The coefficient of D_c at B is an integral multiple of $1/e_B$. Consequently

$$\Delta = R + \Theta_c - \{D_c\} \tag{5.1}$$

is effective: the fractional coefficient being subtracted is at most $(e_B - 1)/e_B$. The pair (X, Δ) is klt. Indeed, on a smooth finite quotient uniformizer $U \rightarrow X$, the log pullback of Δ is the pullback of Θ minus the effective pullback of $\{D_c\}$. This boundary is bounded above by the simple normal crossing boundary $\Theta|_U$ with coefficients less than one. All its log discrepancies are therefore positive. The finite-map discrepancy formula, with ramification included in the displayed canonical identity, gives the same positivity downstairs.

Normal extension from codimension one identifies

$$q_*\mathcal{O}_{\mathcal{X}}(D) = \mathcal{O}_X(\lfloor D_c \rfloor).$$

For completeness, on a finite quotient chart the pushforward is the invariant part of an invertible module on a normal smooth scheme. Sections extend across sets of codimension at least two there and then remain invariant, so this pushforward is the rank-one reflexive sheaf prescribed by its codimension-one orders. Those orders are exactly $\lfloor D_c \rfloor$. Taking invariants is exact in characteristic zero, which also gives

$$H^q(\mathcal{X}, \mathcal{O}_{\mathcal{X}}(D)) = H^q(X, \mathcal{O}_X(\lfloor D_c \rfloor)).$$

The coarse space has quotient singularities and is locally $\mathbb{Q}$ -factorial. Thus $\lfloor D_c \rfloor$ is an integral $\mathbb{Q}$ -Cartier Weil divisor. By Equation (5.1),

$$\lfloor D_c \rfloor - (K_X + \Delta) = D_c - (K_X + R + \Theta_c),$$

which is the nef and big class descended from $D - (K_{\mathcal{X}} + \Theta)$. Kawamata–Viehweg vanishing for a klt pair and an integral $\mathbb{Q}$ -Cartier Weil divisor now applies [12]. It gives the claimed vanishing. This argument requires neither flatness of the coarse morphism nor ampleness in place of nefness and bigness. $\square$

5.2 The adjoint lifting implication

Lemma 5.2 (Adjoint lattice points). *For the orbifold jet data above,*

$$(j, \alpha) \in \Lambda, \quad (j, \alpha) + c_0 \in \text{int } K \implies (j, \alpha) \in \Gamma. \quad (5.2)$$

Proof. The coordinate inequalities of K imply $\alpha_i + 1 > 0$. Since α_i is integral, $\alpha_i \geq 0$. The bounded-slice property and the positivity of the degree coordinate on $K \setminus \{0\}$ give $j + r > 0$. Choose a rational number

$$0 < j_0 < j + r \quad \text{such that} \quad p = (j_0, \alpha + \mathbf{1}) \in \text{int } K.$$

We construct a section of jL with the required Taylor initial.

Replacing finitely many initials by scalar weights. There are finitely many sections whose initial indices span a polyhedral cone containing p in its interior. To see this without assuming finite generation of Γ , choose a small simplex around p inside K and approximate its vertices by points in the cone generated by Γ . Sufficiently close approximations still surround p , and each uses only finitely many semigroup elements. The union of those finite lists has the desired property.

For this list of sections, choose positive primitive integral weights $u_1, \dots, u_N$ such that scalar u -order has exactly their specified Taylor initials, each as a single monomial. Here is the finite-order justification. Let d exceed the total degrees of those initials and the total degree of α . Weights sufficiently close to a common positive value preserve all strict total-degree comparisons through degree d and place every monomial of total degree greater than d above the relevant initials. Among monomials of equal degree at most d , successive sufficiently small rational perturbations of the weights realize the chosen lexicographic order. There are only finitely many such comparisons. We can include all monomials preceding α among them, and then clear denominators and divide by the common divisor. Thus, in addition,

$$\gamma \text{ precedes } \alpha \implies u \cdot \gamma < u \cdot \alpha. \quad (5.3)$$

This handles formal Taylor series as well as polynomial expansions, since positivity of the weights excludes all sufficiently high total degrees at once.

Set

$$p_0 = u \cdot \alpha, \quad U = u \cdot (\alpha + \mathbf{1}) = p_0 + \sum_i u_i.$$

The affine plane given by original degree j_0 and scalar cost U meets the finite cone just chosen in a relative neighborhood of p . Choose finitely many rational points in that neighborhood surrounding p in the plane. Their expressions as nonnegative combinations of the integral cone generators can be taken rational. Clear all denominators simultaneously. Multiplication of the corresponding sections then gives a positive integer a and sections $s_1, \dots, s_t$ of aj_0L with single u -initials $z^{\beta^{(1)}}, \dots, z^{\beta^{(t)}}$, after rescaling, such that

$$u \cdot \beta^{(k)} = aU, \quad \alpha + \mathbf{1} \in \text{relint conv}\{\beta^{(1)}/a, \dots, \beta^{(t)}/a\} \subset \{x : u \cdot x = U\}. \quad (5.4)$$

The relative interior here contains a neighborhood in the entire displayed affine plane. Products preserve single initials, so no claim about generation of the full section ring has entered this construction.

The weighted blowup stack. Use the algebraic equivariant parameters fixed in Section 4. For $k \geq 0$, let I_k be the ideal generated locally by monomials of u -weight at least k . Each I_k contains an ordinary power of the maximal ideal, so its quotient is determined on a finite infinitesimal neighborhood of the chosen residual gerbe. Equivariance descends this quotient to B , and extending its kernel as the unit ideal off the center gives a global coherent ideal. On a chart with a larger finite group, use the translates at all points of the finite orbit. One fixed parameter system defines every I_k , so these ideals form a multiplicative filtration, including after passage to the completed Taylor ring.

The weighted Rees algebra

$$\mathcal{A} = \bigoplus_{k \geq 0} I_k \mathfrak{q}^k$$

is locally generated over the coordinate ring by $z_i \mathfrak{q}^\ell$, $1 \leq \ell \leq u_i$: distribute k among the factors of a monomial of weight at least k , assigning at most u_i to each occurrence of z_i . Define $\pi : \mathcal{Z} \rightarrow B$ as the stack quotient by the grading $\mathbb{G}_m$ of $\mathrm{Spec}_B \mathcal{A} \setminus V(\mathcal{A}_+)$. This global construction commutes with the etale parameter charts and is the identity away from the center, where $I_k = \mathcal{O}_B$.

We identify its local smooth model to retain the exceptional inertia. Put $x_i = z_i \mathfrak{q}^{u_i}$. These elements cover the irrelevant open by their nonvanishing loci, because

$$(z_i \mathfrak{q}^\ell)^{u_i} = x_i^\ell z_i^{u_i - \ell}.$$

On $x_i \neq 0$, the ratio $t = (z_i \mathfrak{q}^{u_i - 1})/x_i$ is regular, including when $u_i = 1$, and $z_j = t^{u_j} x_j$. Thus before taking the original cyclic quotient our model is

$$\left[(\mathbb{A}_{x_1, \dots, x_N, t}^{N+1} \setminus \{x_1 = \dots = x_N = 0\}) / \mathbb{G}_m \right], \quad \rho \cdot (x_i, t) = (\rho^{u_i} x_i, \rho^{-1} t).$$

The original μ_m acts by its characters on x_i and trivially on t . The atlas is smooth and all stabilizers are finite. The exceptional Cartier divisor E is $t = 0$; it is $[\mathbb{P}(u_1, \dots, u_N)/\mu_m]$, with any generic stabilizer retained. The coarse modification is the relative projective weighted Rees blowup. A sufficiently divisible $\mathcal{O}_{\mathcal{Z}}(-qE)$ descends to its relatively ample tautological line bundle. Since the coarse space of B is projective, so is that of $\mathcal{Z}$.

The Jacobian calculation in these coordinates gives

$$K_{\mathcal{Z}} + E = \pi^* K_B + \left(\sum_i u_i \right) E. \quad (5.5)$$

Indeed $\bigwedge_i dz_i$ has order $\sum_i u_i - 1$ along $t = 0$ on each uniformizer. Keeping the stack structures makes this the Cartier divisor identity asserted in Equation (5.5).

Define the line bundle divisors

$$P_0 = \pi^*(jL) - p_0 E, \quad H_0 = \pi^*(aj_0 L) - aUE.$$

The canonical shift cancels the exceptional order because $U = p_0 + \sum_i u_i$. Together with $-K_B = rL$, this gives

$$P_0 - (K_{\mathcal{Z}} + E) - \frac{1}{a} H_0 = (j + r - j_0) \pi^* L. \quad (5.6)$$

Its right side is nef and big, with strictly positive coefficient; this is the positivity needed to lift from E . After their common exceptional order is removed, the sections $s_1, \dots, s_t$ define sections of H_0 . Let $\mathfrak{b}$ be their base ideal. It does not vanish identically on E , and its restriction there is generated locally by the indicated monomial initials.

Multiplier-ideal membership on the exceptional divisor. The congruence $(j, \alpha) \in \Lambda$ says that z^α has exactly the character required to be a global section of $P_0|_E$. Its scalar degree is p_0 , and $\mathcal{O}_E(-E) = \mathcal{O}_E(1)$ in the weighted projective convention. We claim that

$$z^\alpha \in H^0(E, P_0|_E \otimes \mathcal{J}_E((\mathfrak{b}|_E)^{1/a})). \quad (5.7)$$

On the chart where $x_k \neq 0$, pass to the smooth finite uniformizer that sets $x_k = 1$. More precisely the chart is $[\mathbb{A}^{N-1}/H_k]$, where

$$H_k = \{(\rho, \zeta) \in \mathbb{G}_m \times \mu_m : \rho^{u_k} \zeta^{b_k} = 1\}.$$

This is a finite extension of μ_m by μ_{u_k} , acting on the remaining coordinates by $x_i \mapsto \rho^{u_i} \zeta^{b_i} x_i$. It need not be a direct product or act effectively. On the underlying affine uniformizer, $P_0|_E$ and $H_0|_E$ are trivial line bundles with the respective H_k -characters $\rho^{p_0} \zeta^j$ and $\rho^{aU} \zeta^{aj_0}$. Thus the generators of the restricted ideal become the ordinary monomials $x_{\text{rem}}^{\beta_{\text{rem}}^{(i)}}$, and the proposed section becomes $x_{\text{rem}}^{\alpha_{\text{rem}}}$. For example, on H_k ,

$$\prod_{i \neq k} (\rho^{u_i} \zeta^{b_i})^{\alpha_i} = \rho^{p_0} \zeta^j,$$

by the defining relation of H_k and the congruence for α . There are no fractional exponents: the root needed to set $x_k = 1$ has already been taken in the smooth atlas. Any generic stabilizer of E is retained in H_k and imposes exactly these character conditions, without changing the smooth-atlas multiplier formula. Projection from the plane $u \cdot x = U$ by dropping coordinate k is an affine isomorphism. Hence Equation (5.4) places $\alpha_{\text{rem}} + \mathbf{1}$ in the ordinary interior of the convex hull of the projected scaled generators, and therefore in the interior of their scaled Newton polyhedron.

The monomial multiplier-ideal formula says that x^ν belongs to $\mathcal{J}(I^c)$ precisely when $\nu + \mathbf{1}$ is in the interior of c times the Newton polyhedron of I [17]. Its hypotheses hold on this smooth affine chart. Applying it with $c = 1/a$ proves the claimed membership on every uniformizer. The ideal and the section are equivariant, so these memberships descend and prove Equation (5.7). No ordinary smoothness of the coarse exceptional divisor is being assumed.

Adjoint vanishing and restriction. We apply the positivity in Equation (5.6) on a projective log resolution $\sigma : \mathcal{Y} \rightarrow \mathcal{Z}$ of E and $\mathfrak{b}$, isomorphic at the generic point of E . We use smooth stacks throughout: functorial log resolution in smooth etale charts [23, Theorems 35–36] descends. Arrange that the total boundary, including the strict transform $\tilde{E}$, has simple normal crossings, and that the restricted morphism $\tau : \tilde{E} \rightarrow E$ resolves $\mathfrak{b}|_E$. Write

$$\mathfrak{b}\mathcal{O}_{\mathcal{Y}} = \mathcal{O}_{\mathcal{Y}}(-G_0).$$

The effective divisor G_0 does not contain $\tilde{E}$, and $\sigma^*H_0 - G_0$ is globally generated by the pulled-back sections.

Consider the integral line bundle divisor

$$A_{\mathcal{Y}} = K_{\mathcal{Y}} + \sigma^*(P_0 - K_{\mathcal{Z}} - E) - \lfloor G_0/a \rfloor.$$

Its difference from $K_{\mathcal{Y}} + \{G_0/a\}$ is

$$(j + r - j_0)\sigma^*\pi^*L + \frac{1}{a}(\sigma^*H_0 - G_0). \quad (5.8)$$

The first term is nef and big and the second is nef. The fractional boundary has simple normal crossings and coefficients less than one. Lemma 5.1 gives $H^1(\mathcal{Y}, \mathcal{O}_{\mathcal{Y}}(A_{\mathcal{Y}})) = 0$. The divisor restriction sequence therefore gives a surjection

$$H^0(\mathcal{Y}, A_{\mathcal{Y}} + \tilde{E}) \longrightarrow H^0(\tilde{E}, (A_{\mathcal{Y}} + \tilde{E})|_{\tilde{E}}). \quad (5.9)$$

Adjunction identifies the bundle on the right with

$$\tau^*(P_0|_E) + K_{\tilde{E}} - \tau^*K_E - \lfloor (G_0|_{\tilde{E}})/a \rfloor. \quad (5.10)$$

Here round-down commutes with restriction. Indeed, simple normal crossings of $\tilde{E} + \text{Supp } G_0$ prevents distinct components of G_0 from sharing a divisor on $\tilde{E}$. If an individual intersection is disconnected, its coefficient is the same on each component, so the equality still holds. Moreover $G_0|_{\tilde{E}}$ is the divisor of the pulled-back restricted ideal. Thus Equation (5.10) is precisely the bundle in the log-resolution definition of the multiplier ideal appearing in Equation (5.7). That membership supplies a section of Equation (5.10) whose rational section is the pullback of z^α . Surjectivity in Equation (5.9) lifts it to $\mathcal{Y}$.

Regular pushdown and recovery of the initial. The difference between the lifting bundle and σ^*P_0 is

$$K_{\mathcal{Y}} - \sigma^*K_{\mathcal{Z}} - \sigma^*E + \tilde{E} - \lfloor G_0/a \rfloor. \quad (5.11)$$

The part preceding the last term is effective and exceptional over $\mathcal{Z}$. This can be checked during a resolution by smooth blowups: a center of codimension $c \geq 2$ contributes $c - 1$, or $c - 2$ when contained in the strict transform of E , in addition to pullbacks of already effective exceptional terms. The same calculation holds in the etale smooth charts. The final term in Equation (5.11) is the negative of an effective divisor. It follows that the lifted rational section has no pole along any nonexceptional prime divisor. It pushes down to a regular section of P_0 on the normal stack $\mathcal{Z}$. Its restriction to E agrees generically with z^α , and hence agrees everywhere, since both are regular sections on the integral smooth stack E .

Finally $p_0 \geq 0$, so P_0 is a subsheaf of $\pi^*(jL)$. Normal extension across the center, or the local weighted blowup charts, gives $\pi_*\mathcal{O}_{\mathcal{Z}} = \mathcal{O}_B$. Pushing down thus produces a section of jL . The exceptional order and the specified restriction say exactly that its u -initial is z^α , with no term of smaller u -cost. Equation (5.3) then excludes every Taylor monomial preceding α in the original order. This proves Equation (5.2). $\square$

5.3 Integral support forms and saturation

We now use the interior conclusion from the preceding section. The next argument explains why the adjoint implication forces a polyhedral cone, rather than merely imposing a condition on facets that might happen to be rational.

Theorem 5.3 (The saturated jet semigroup). *Suppose $c_0 \in \text{int } K$. Then the following statements hold:*

1. K is a rational polyhedral cone.
2. Every facet has a primitive nonnegative support form $\eta \in \Lambda^*$ satisfying $\eta(c_0) = 1$.
3. The semigroup and its interior lattice points satisfy

$$\Gamma = K \cap \Lambda, \quad \text{int } K \cap \Lambda = c_0 + \Gamma. \quad (5.12)$$

In particular, Γ is a finitely generated saturated affine semigroup with group Λ . These conclusions apply to every sufficiently far grading and chosen point covered by Theorem 4.8, including its exact rational case.

Proof. Fix such jet data, with $c_0 \in \text{int } K \cap \Lambda$. Put $\varphi(x) = (r/N)f(x)$, so the epigraph of φ uses the original degree coordinate j , rather than $D_0 = Nj/r$. On the open positive orthant, φ is a finite convex function and is locally Lipschitz.

A support at a differentiability point has height one. Let x be a differentiability point of φ in the positive orthant, and let $y = (\varphi(x), x) \in \partial K$. Homogeneity gives $\varphi(x) = \nabla \varphi(x) \cdot x$. The unique supporting normal ray at y is therefore generated by

$$\eta(j', x') = j' - \nabla \varphi(x) \cdot x',$$

and $\eta(c_0) > 0$. Suppose either that this ray is irrational relative to Λ , or that its primitive integral generator has value greater than one at c_0 . In both cases there exists $w \in \Lambda$ with

$$-\eta(c_0) < \eta(w) < 0. \quad (5.13)$$

For an integral primitive form of height at least two, take an element of value -1 . For an irrational ray, the subgroup $\eta(\Lambda) \subset \mathbb{R}$ is dense: any nondense additive subgroup of $\mathbb{R}$ is discrete, and discreteness here would make a positive multiple of η primitive integral. This proves the assertion in the irrational case as well.

There are integers $k_\nu \rightarrow \infty$ and lattice points $l_\nu \in \Lambda$ such that

$$e_\nu = l_\nu - k_\nu y \rightarrow 0. \quad (5.14)$$

This is recurrence of the multiples of y in the compact torus $\mathbb{R}^{N+1}/\Lambda$. It also follows directly by the pigeonhole principle applied to successively finer finite partitions of a lattice fundamental domain. If the resulting return times stay bounded, one of them is an exact period and its unbounded multiples give Equation (5.14) instead.

Now $l_\nu + w$ is outside K for large ν , since its η -value tends to the negative number $\eta(w)$. In contrast,

$$l_\nu + w + c_0 = k_\nu \left(y + \frac{w + c_0 + e_\nu}{k_\nu} \right) \in \text{int } K \quad (5.15)$$

for large ν . To justify this strict inclusion, set $v = w + c_0$, so $\eta(v) > 0$ by Equation (5.13). Differentiability gives, uniformly for v' in a small neighborhood of v ,

$$\varphi(x + tv'_x) = \varphi(x) + t \nabla \varphi(x) \cdot v'_x + o(t).$$

Thus the j -coordinate of $y + tv'$ exceeds $\varphi(x + tv'_x)$ for every sufficiently small positive t ; the exponent coordinates remain positive as well. Apply this with $t = 1/k_\nu$ and $v' = v + e_\nu$ to obtain Equation (5.15). Lemma 5.2, applied to the lattice point $l_\nu + w$, would then put this point in $\Gamma \subset K$, a contradiction. Every supporting ray at a differentiability point therefore has a primitive integral form of height one at c_0 .

Only finitely many support forms occur. Choose $\epsilon > 0$ such that the Euclidean ball of radius ϵ about c_0 is contained in K . If a nonnegative support form has $\eta(c_0) = 1$, testing it on this ball gives $\|\eta\| \leq 1/\epsilon$. Since Λ^* is a lattice, only finitely many primitive integral forms satisfy this bound. Denote the forms obtained at differentiability points by $\eta_1, \dots, \eta_t$.

These forms, together with the coordinate inequalities, define K . Indeed, convex functions are differentiable almost everywhere on their open domain. At any given positive x_0 , take

differentiability points $x_\nu \rightarrow x_0$. Local Lipschitz continuity bounds their gradients, and therefore

$$\varphi(x_\nu) + \nabla \varphi(x_\nu) \cdot (x_0 - x_\nu) \rightarrow \varphi(x_0).$$

Each expression is the value at x_0 of a supporting linear function. Hence the supports at these differentiability points recover the epigraph on the positive orthant. Since only the finite list above occurs, K and

$$K' = \{(j, x) : x_i \geq 0 \text{ for all } i, \eta_a(j, x) \geq 0 \text{ for all } a\}$$

agree there. We already have $K \subset K'$. Conversely, for $q \in K'$, the point $q + tc_0$ lies in K' and has positive exponent coordinates whenever $t > 0$. It is thus in K , and closedness gives $q \in K$ on letting $t \downarrow 0$. This proves $K = K'$ and rational polyhedrality.

The coordinate form $(j, x) \mapsto x_i$ is primitive on Λ : one can prescribe $x = e_i$ and choose j in the requisite congruence class. Its value on c_0 is one. All other defining forms also have height one. Every facet of the full-dimensional polyhedral cone is cut out by a form from an irredundant sublist, so its primitive support has the asserted height.

Saturation and the interior translation. If $q \in K \cap \Lambda$, then $q + c_0 \in \text{int } K$: translate a small ball about c_0 by q and use addition in the convex cone. Lemma 5.2 gives $q \in \Gamma$, proving the first identity in Equation (5.12). If $q \in \text{int } K \cap \Lambda$, each primitive facet form takes a positive integral value on q , hence a value at least one. Since its value on c_0 is one, all facet inequalities hold for $q - c_0$. Thus $q - c_0 \in K \cap \Lambda = \Gamma$. The reverse inclusion $c_0 + \Gamma \subset \text{int } K \cap \Lambda$ follows from the same ball-translation argument. This proves the second identity.

Finally, triangulate the rational polyhedral cone into finitely many rational simplicial cones and choose lattice generators for their rays. A lattice point in any such cone is a nonnegative integral combination of its ray generators plus a lattice point in their bounded half-open parallelepiped. There are only finitely many such remainder points. Taking their union over the triangulation proves finite generation of $K \cap \Lambda$. If a positive integral multiple of $q \in \Lambda$ lies in this semigroup, then $q \in K$ by homogeneity, so q already lies in it; this is saturation. The group is Λ by Proposition 4.1. $\square$

6 Two filtrations and a second stabilizer

Fix one of the gradings and points furnished by Theorem 5.3. We retain the original degree j and put $n = N + 1$, with $N \geq 4$. Thus $S = \mathbb{C}[C]$ is a positively graded normal Gorenstein domain, its vertex o is singular, and its canonical module has a homogeneous generator of original degree r . The Taylor semigroup and its lattice satisfy

$$\Gamma = K \cap \Lambda, \quad \text{int}(K) \cap \Lambda = c_0 + \Gamma, \quad c_0 = (r, 1, \dots, 1). \quad (6.1)$$

Here K is rational polyhedral. We also use the finite Taylor-order bound and the eventual realization of compatible jets from Proposition 4.1. The only volume hypothesis in this section is

$$\widehat{\text{vol}}(o, C) \geq 2N^n. \quad (6.2)$$

The first filtration separates total Taylor order from original degree. We then assign positive cost to the constant Taylor direction as well. The resulting second graded ring has a nontrivial stabilizer, which will supply the same-dimensional slice in Section 7.

6.1 The first associated graded ring

For an original-homogeneous section $s \in S_j$, let $T(s)$ be its total Taylor order at the chosen orbifold point, computed in the fixed smooth uniformizing chart and local frame. Extend the filtration by direct sums in original degree, and write

$$\bar{S} = \text{gr}_T S.$$

Its bidegrees will always mean original degree and T -degree, in that order. If τ is a book-keeping variable for original degree, then $\bar{S}$ is the subalgebra of $\mathbb{C}[\tau, z_1, \dots, z_N]$ consisting of homogeneous Taylor forms $\tau^j P(z)$ obtained from sections. Each such form satisfies the stabilizer congruence. Taking its further monomial initial with the fixed lexicographic refinement gives the algebra $\mathbb{C}[\Gamma]$.

The saturated monomial algebra will determine normality and the canonical degree of $\bar{S}$. A second consequence of saturation is a section whose Taylor form is the constant τ^m . Inverting that section recovers every compatible Taylor monomial; this will determine the Hilbert series of its zero fiber and give a basis for the next filtration.

Lemma 6.1 (The Taylor graded ring). *The algebra $\bar{S}$ is finitely generated, normal, Cohen–Macaulay, and Gorenstein. Its canonical module has a homogeneous generator of bidegree (r, N) . There is a section $v \in S_m$ whose initial $\bar{v}$ is τ^m . In particular $T(v) = 0$, and*

$$\bar{S}[\bar{v}^{-1}] \simeq \mathbb{C}[\bar{v}, \bar{v}^{-1}, u_1, \dots, u_N], \quad T(u_i) = 1, \quad T(\bar{v}) = 0. \quad (6.3)$$

The ring $D = \bar{S}/(\bar{v})$ is an equidimensional Cohen–Macaulay Gorenstein ring of dimension N , with

$$\text{Hilb}_T D(z) = (1 - z)^{-N}, \quad \deg \omega_D = (r - m, N). \quad (6.4)$$

Its T -degree-zero piece is $\mathbb{C}$. A bihomogeneous vector-space basis of D can be lifted to sections $e_i \in S$ such that, on putting $T_i = T(e_i)$,

$$\{v^a e_i : a \in \mathbb{N}\} \quad (6.5)$$

is a vector-space basis of S , compatible with original degree and T -order. The unique vector of this basis with $a + T_i = 0$ is 1.

Proof. We first transfer the structure of the monomial algebra to $\bar{S}$. By Equation (6.1), Γ is a finitely generated normal affine semigroup. Hochster’s theorem makes $\mathbb{C}[\Gamma]$ Cohen–Macaulay, and the canonical-module formula for normal affine semigroup rings identifies its canonical module with the monomial ideal of interior lattice points; see [16] and [47, Theorem 6.7 and its following normal-monoid reformulation]. The second identity in Equation (6.1) makes this ideal principal, generated in multidegree $(r, 1, \dots, 1)$. Thus its bidegree is (r, N) . The interior-translation identity, rather than saturation alone, is essential here.

Choose a finite set of generators of Γ and bihomogeneous lifts in $\bar{S}$ of their monomials. We lift generation by degree-wise subduction, in the terminology of Khovanskii bases [20, Algorithm 2.11 and Proposition 2.13]. Subtracting products of the lifts cancels any selected initial monomial. Within a fixed original degree there are only finitely many Taylor orders, and within each order only finitely many monomials. Consequently this subtraction terminates and the chosen lifts generate $\bar{S}$.

The same finite list gives a one-parameter flat monomial degeneration. Choose positive integral costs on the z_i that preserve the selected monomial initial of every one of these finitely many lifts. Such costs exist by approximating the specified lexicographic refinement on this finite list. The scalar initial algebra contains $\mathbb{C}[\Gamma]$. In each original bidegree its dimension is the dimension of $\bar{S}$ in that bidegree, which is also the number of the corresponding points

of Γ . The inclusion is therefore an equality in every bidegree. Lifting the generators again, with the same degreewise terminating subtraction, shows that the extended Rees algebra for this scalar filtration is finite type. As a subalgebra of a Laurent-polynomial extension it is torsion-free, hence flat, over its parameter line. Its zero fiber is $\mathbb{C}[\Gamma]$ and its nonzero fibers are $\bar{S}$.

Write R for this extended Rees algebra and λ for its parameter. It retains original degree and T -degree, and has the additional scalar filtration grading, with scalar weight -1 on λ . A sufficiently large multiple of original degree minus scalar weight gives a positive grading on R , including positive degree for λ . At its vertex, the local ring A has a normal Gorenstein quotient $A/(\lambda)$. Lifting through the nonzerodivisor λ makes A Cohen–Macaulay and Gorenstein; for the latter equivalence see [48, Tag 0BJJ].

The local ring A is also normal. Here is a direct verification using $R_1 + S_2$. It is already Cohen–Macaulay. Let P be a height-one prime of A . If $\lambda \in P$, reducedness of $A/(\lambda)$ makes $A_P/(\lambda)$ a field, so A_P is regular. Otherwise choose Q minimal over (P, λ) . Since A is a local domain essentially of finite type over $\mathbb{C}$, the dimension formula and the principal ideal theorem give $\text{ht } Q \leq 2$. The normal ring $A_Q/(\lambda)$ has dimension at most one and is regular. Lifting a regular system of parameters together with λ shows that A_Q is regular, and then so is its localization A_P .

The normal and Gorenstein loci of $\text{Spec } R$ are open, contain its vertex, and are invariant under the positive grading action. Every orbit contracts to that vertex; hence R is normal and Gorenstein everywhere. Its generic fiber, a localization of R , is $\bar{S} \otimes_{\mathbb{C}} \mathbb{C}(\lambda)$ and is normal. Normality descends under this faithfully flat field extension, proving normality of $\bar{S}$. Quotienting R by $\lambda - 1$ proves the Cohen–Macaulay and Gorenstein assertions for $\bar{S}$ as well.

Adjunction tracks the canonical bidegree. The parameter λ has original degree and T -degree both zero, so it contributes no shift in these two degrees. Choose a lift of the central canonical generator homogeneous also for the scalar grading, with its adjunction shift. Its cokernel in the canonical module of R is equal to its product by λ . Graded Nakayama for the positive combined grading therefore makes the cokernel zero. Restriction to $\lambda = 1$ forgets the scalar grading but preserves original degree and T -degree. Thus the canonical generator of $\bar{S}$ has bidegree (r, N) .

We next use the constant Taylor direction to describe $\bar{S}$ over a polynomial parameter. The vector $(m, 0)$ lies on the constant ray of K and belongs to Λ . Equation (6.1) provides a section with this leading index. Rescale its value to obtain $\bar{v} = \tau^m$. Choose integer representatives b_i of the stabilizer characters. Eventual compatible jet separation shows that, after inverting τ^m , every monomial $\tau^j z^\gamma$ with

$$j - \sum_i b_i \gamma_i \in m\mathbb{Z}$$

occurs. Set $u_i = \tau^{b_i} z_i$ in that localization. The displayed congruence expresses each allowed monomial uniquely as a Laurent power of $\bar{v}$ times a monomial in the u_i . This proves Equation (6.3), including its T -grading. It also shows directly that there is no residual finite quotient in a nonzero $\bar{v}$ fiber.

For each $q \geq 0$, consider the T -degree- q piece of $\bar{S}$ as a module over $\mathbb{C}[\bar{v}]$. It is torsion-free. It is also finite: in each original-degree residue class modulo m , multiplication by $\bar{v}$ identifies its pieces with an increasing sequence of subspaces of the finite-dimensional space of compatible degree- q Taylor forms. These subspaces are eventually the full compatible space by jet separation. Thus there are only finitely many necessary module generators. This module is consequently free over the principal ideal domain $\mathbb{C}[\bar{v}]$. Equation (6.3) shows that its rank is

$$\binom{q + N - 1}{N - 1}.$$

Reduction modulo $\bar{v}$ consequently has exactly this dimension in T -degree q , proving the Hilbert series in Equation (6.4).

Since $\bar{v}$ is a nonzerodivisor in the normal Cohen–Macaulay domain $\bar{S}$, its quotient is equidimensional of dimension N and Cohen–Macaulay. Adjunction makes the quotient Gorenstein and subtracts $(m, 0)$ from the canonical degree. The Hilbert series shows that the T -degree-zero piece of D is $\mathbb{C}$.

It remains to obtain a basis of S compatible with the filtration. Lift a bihomogeneous basis of D to $\bar{S}$. These lifts are a $\mathbb{C}[\bar{v}]$ -basis: generation follows by subtracting modulo $\bar{v}$ and descending in original degree, and independence follows by reducing a putative relation modulo $\bar{v}$ and successively removing its smallest power. Now lift them through the T -filtration to original-homogeneous sections e_i of the indicated orders T_i . Their products with powers of v have the just-constructed basis as initials. Terminating cancellation in original degree proves Equation (6.5). Choose $e_i = 1$ for the sole basis vector of T -degree zero; all other T_i are positive. $\square$

6.2 Canonical forms on extended Rees spaces

We make the discrepancy comparison used for both filtrations explicit. If F is a nonnegative decreasing multiplicative integral filtration on S , set $F_p S = S$ for $p \leq 0$ and use the extended Rees convention

$$R_F = \sum_{p \in \mathbb{Z}} \lambda^{-p} F_p S \subset S[\lambda, \lambda^{-1}]. \quad (6.6)$$

The filtration action has weight zero on S and weight -1 on λ . In particular, $R_F/(\lambda) = \text{gr}_F S$ and $R_F[\lambda^{-1}] = S[\lambda, \lambda^{-1}]$. The inclusion $S[\lambda] \subset R_F$ gives a birational morphism $\text{Spec } R_F \rightarrow C \times \mathbb{A}^1$.

Lemma 6.2 (Rees discrepancy comparison). *Suppose F respects original degree, has finite length in each original homogeneous piece, and $\text{gr}_F S$ is finitely generated. Suppose also that $\mathfrak{X} = \text{Spec } R_F$ is normal and Gorenstein, and that $\text{gr}_F S$ is Cohen–Macaulay and Gorenstein with a canonical generator of bidegree (r, Q) . Then R_F is finite type and its absolute canonical module has a homogeneous generator of bidegree $(r, Q - 1)$, which on $\lambda \neq 0$ is*

$$\Omega_{\mathfrak{X}} = c \lambda^{-Q} \Omega_C \wedge d\lambda, \quad c \in \mathbb{C}^*. \quad (6.7)$$

For a quasi-monomial valuation V centered on $\mathfrak{X}$ over $\lambda = 0$, put $\ell = V(\lambda) > 0$ and $w = V|_{\mathbb{C}(C)}$. If w is centered at o , then

$$A_C(w) \leq A_{\mathfrak{X}}(V) + (Q - 1)\ell. \quad (6.8)$$

The same inequality holds whenever the discrepancies involved are finite and defined by valuation retractions.

Proof. Lift homogeneous generators of $\text{gr}_F S$ to $s_i \in S$, of filtration orders p_i . Cancellation of initial forms, terminating within each original degree, shows that R_F is generated by λ and $\lambda^{-p_i} s_i$. Indeed, an element of $F_p S$ has an expression as a sum of monomials in the s_i whose filtration orders are at least p ; multiplication by λ^{-p} then introduces only nonnegative extra powers of λ . Taking a sufficiently large multiple of original degree minus filtration degree makes every one of these generators positive, with $\deg \lambda = 1$.

Cartier adjunction [48, Tag 0B4A] for the nonzerodivisor λ identifies the reduction of the canonical module of R_F with that of $\text{gr}_F S$, with the shift coming from $d\lambda$. Since λ has filtration weight -1 , a lift of the specified central generator has bidegree $(r, Q - 1)$. This lift generates the whole canonical module: its cokernel is equal to its product by λ , and a finite

module bounded below in the positive combined grading cannot have that property unless it is zero. As the canonical module is torsion-free of rank one over the domain R_F , the resulting generator is free.

On $\lambda \neq 0$, both this generator and $\Omega_C \wedge d\lambda$ generate the same canonical module. Their ratio is a unit of $S[\lambda, \lambda^{-1}]$. The units of the positively graded domain S are the nonzero constants, so such a unit is $c\lambda^a$. The filtration weight of $\Omega_C \wedge d\lambda$ is -1 , and the weight of the lifted generator is $Q - 1$. It follows that $a = -Q$, proving Equation (6.7). Computing on a common smooth birational model gives

$$A_{C \times \mathbb{A}^1}(V) = A_{\mathfrak{X}}(V) + Q\ell. \quad (6.9)$$

This computation uses rational top forms and does not require the Rees morphism itself to be proper.

On the other hand,

$$A_{C \times \mathbb{A}^1}(V) \geq A_C(w) + \ell. \quad (6.10)$$

To see this directly, take any smooth proper birational log model of C used to retract w , and take its product with the parameter line. Include $\lambda = 0$ among the simple-normal-crossing divisors. The monomial retraction of V on this product has discrepancy equal to the discrepancy of the corresponding retraction of w plus ℓ . Log discrepancy dominates its value on each such retraction [18, Corollary 5.4 and Remark 5.6]; taking the supremum over the models of C proves Equation (6.10). Combining it with Equation (6.9) proves the lemma. $\square$

6.3 Reducedness and the second filtration

Lemma 6.3 (Reducedness of the quotient). *Under Equation (6.2), D is reduced.*

Proof. The T -Rees algebra is finite type by Lemma 6.1 and the generator-lifting argument in Lemma 6.2. Its nonzero fibers are S , and its central fiber $\bar{S}$ is normal, Cohen–Macaulay, and Gorenstein. The regular parameter and these fiber properties imply that the total space is Cohen–Macaulay and Gorenstein. It is normal as well: away from the central fiber it is $C \times \mathbb{G}_m$, and at every codimension-one point of the central fiber the quotient by λ is a field. Thus it is regular in codimension one and satisfies Serre’s condition S_2 .

Let H be an irreducible component of the divisor $\bar{v} = 0$ in $\text{Spec } \bar{S}$, and write $\nu = \text{ord}_H(\bar{v}) \geq 1$. At its generic point the normal central fiber is a discrete valuation ring. The total space is therefore regular there, of local dimension two, with parameters λ and a lift t of a central uniformizer. In this local ring

$$v = at^\nu + \lambda b$$

for a unit a and a regular element b . Take the monomial valuation with costs $V(\lambda) = 1$ and $V(t) = 1/\nu$. Its discrepancy on the total space is $1 + 1/\nu$. For its restriction w to $\mathbb{C}(C)$ one has

$$w(v) \geq 1, \quad w(e_i) \geq T_i,$$

because $\lambda^{-T_i}e_i$ is regular on the Rees space. Hence $w(v^a e_i) \geq a + T_i$.

By Lemma 6.1, every nonconstant original-homogeneous basis vector has a strictly positive lower bound $a + T_i$. Thus w is centered exactly at o . The series counting the basis with these costs is

$$\sum_{a,i} z^{a+T_i} = (1-z)^{-1} \text{Hilb}_T D(z) = (1-z)^{-n}.$$

The vectors of cost below k consequently number $k^n/n! + O(k^{n-1})$. They span the valuation quotient, so

$$\text{vol}(w) \leq 1. \quad (6.11)$$

These finite-colength quotients of S are supported at o ; their lengths agree with those after localization at the vertex.

Apply Lemma 6.2 with $Q = N$ and $\ell = 1$. It gives

$$A_C(w) \leq N + 1/\nu.$$

If $\nu \geq 2$, Equation (6.11) yields

$$\widehat{\text{vol}}(o, C) \leq (N + 1/2)^{N+1} < 2N^{N+1}.$$

For the strict inequality, use

$$(N + 1) \log \left(1 + \frac{1}{2N} \right) < \frac{N + 1}{2N} \leq \frac{5}{8} < \log 2, \quad N \geq 4.$$

This contradicts Equation (6.2). All the multiplicities of $\bar{v}$ are therefore one. Its quotient D is generically reduced, and it is Cohen–Macaulay, hence satisfies S_1 . The reducedness criterion $R_0 + S_1$ proves the assertion; see [48, Tag 033P]. $\square$

Define a second filtration by assigning the basis in Equation (6.5) the costs

$$W(v^a e_i) = a + T_i \tag{6.12}$$

and taking spans of basis vectors of cost at least the indicated level. Unlike T , this filtration is positive on every nonconstant original-homogeneous vector.

There is also a description independent of the lifted basis. Write $T_q S = \{s : T(s) \geq q\}$, with $T_q S = S$ for $q \leq 0$. For the distinguished section v fixed above, one has

$$W_p S = \sum_{a=0}^p v^a T_{p-a} S \quad (p \geq 0). \tag{6.13}$$

Indeed, every basis term $v^b e_i$ on the right has $b + T_i \geq p$. Conversely, for such a term choose $a = \min\{p, b\}$; then $v^{b-a} e_i \in T_{p-a} S$. Thus changing the compatible lifts e_i does not change W . The multiplicativity of T also makes the right side of Equation (6.13) multiplicative. The remaining point is to identify the associated graded algebra, not merely its Hilbert function.

Proposition 6.4 (The two-filtration structure). *The filtration W is multiplicative, with*

$$G = \text{gr}_W S \simeq D[\mathbf{v}], \quad \deg_W \mathbf{v} = 1, \quad \deg_W |_D = T. \tag{6.14}$$

Here $\mathbf{v}$ is a regular polynomial variable of original degree m . Moreover

$$\text{Hilb}_W G(z) = (1 - z)^{-n}, \quad \deg \omega_G = (r, n). \tag{6.15}$$

The W -Rees space is a finite-type normal Gorenstein variety. Its canonical generator has bidegree $(r, n - 1)$ and, on $\lambda \neq 0$, is $c\lambda^{-n}\Omega_C \wedge d\lambda$.

Proof. Expand a product $e_i e_j$ in the basis $v^a e_k$. Compatibility of that basis with total Taylor initials shows that every nonzero term has $T_k \geq T_i + T_j$. Its W -cost $a + T_k$ is therefore at least $T_i + T_j$. Terms with equality have precisely $a = 0$ and $T_k = T_i + T_j$; their images are the product of the images of e_i, e_j in $D = \bar{S}/(\bar{v})$. Factoring out the displayed powers of v proves multiplicativity and the algebra isomorphism in Equation (6.14). In particular the variable $\mathbf{v}$ is regular even if D has several components.

The Hilbert series follows from Equation (6.4). Adjoining the polynomial variable of bidegree $(m, 1)$ changes the canonical degree $(r - m, N)$ to (r, n) . Thus G is reduced, Cohen–Macaulay, and Gorenstein. The W -filtration has finite length in fixed original degree, as is

clear from its basis description; lifting finitely many generators of G proves finite generation of its Rees algebra.

The nonzero fibers of this Rees family are S , and its central fiber is G . Consequently the total space is Cohen–Macaulay and Gorenstein. For normality, away from $\lambda = 0$ it is $C \times \mathbb{G}_m$. A codimension-one point on $\lambda = 0$ is the generic point of a central component. Since G is reduced, its local ring there is a field, and the total local ring has maximal ideal generated by the nonzerodivisor λ . It is regular. The total space satisfies R_1 and S_2 , hence is normal. Notice that normality of D or of G is not required in this argument. Finally Lemma 6.2, with $Q = n$, gives the canonical generator and top-form identity. $\square$

Example: the quadratic cone. The two stabilizers and filtrations can already differ for an ordinary double point. Let

$$S = \mathbb{C}[a, z_1, \dots, z_N, b]/(ab - q(z)), \quad q(z) = \sum_{i=1}^N z_i^2,$$

with all original degrees equal to one. Here $r = N$ and $m = 1$. At the quotient point $[a : z : b] = [1 : 0 : 0]$, the chart $a = 1$ has $b = q(z)$. Taking $v = a$, the three degrees are

	a	z_i	b
original degree	1	1	1
T	0	1	2
W	1	1	2

The first Taylor graded ring has the same presentation as S , and $D = \mathbb{C}[z_1, \dots, z_N, b]/(q(z))$. The family $\text{Spec } \bar{S} \rightarrow \mathbb{A}_t^1$ defined by $t = \bar{v} = a$ has invariant parameter for the T -action. After inverting t , $b = q(z)/a$ gives the polynomial family of Equation (6.3). With $\mathbf{a} = \lambda^{-1}a$, $\mathbf{z}_i = \lambda^{-1}z_i$, and $\mathbf{b} = \lambda^{-2}b$, the second Rees space is

$$\mathfrak{X} = \{\lambda \mathbf{a} \mathbf{b} - q(\mathbf{z}) = 0\}.$$

Its action has weights $(-1, 1, 1, 2)$ on $(\lambda, \mathbf{a}, \mathbf{z}_i, \mathbf{b})$. The central point with $\mathbf{b} = 1$ and all other coordinates zero therefore has stabilizer $h = 2$. Its transverse slice

$$Y = \{\mathbf{b} = 1\} = \{\lambda \mathbf{a} - q(\mathbf{z}) = 0\}$$

is again an ordinary double point of dimension $n = N + 1$. Thus the original stabilizer $m = 1$ and the new stabilizer $h = 2$ play different roles, and taking this slice preserves the original dimension.

6.4 A nontrivial stabilizer on the new central ring

Lemma 6.5 (Free gradings and polynomial rings). *Let E be a finitely generated positively graded reduced equidimensional Cohen–Macaulay $\mathbb{C}$ -algebra of dimension $N \geq 2$, with $E_0 = \mathbb{C}$ and $\text{Hilb } E(z) = (1 - z)^{-N}$. If the grading action has trivial stabilizer at every nonvertex closed point of $\text{Spec } E$, then E is a polynomial ring on N generators of degree one.*

Proof. We first justify that weight one defines an honest line bundle on $Z = \text{Proj } E$. Choose positive homogeneous algebra generators a_j , with degrees w_j . At a nonvertex closed point q , its grading stabilizer is

$$\mu_{\gcd\{w_j : a_j(q) \neq 0\}}.$$

By hypothesis this gcd is one. Invert the product f of those active generators. Bézout's identity supplies integers c_j with $\sum_j c_j w_j = 1$, so $u = \prod_j a_j^{c_j}$ is an invertible homogeneous element of degree one in E_f . Consequently

$$E_f = (E_f)_0[u, u^{-1}]. \quad (6.16)$$

These open sets cover the puncture: they contain all its closed points, and a finite-type scheme over $\mathbb{C}$ is Jacobson. Their degree-zero charts cover Z . Equation (6.16) gives a Zariski $\mathbb{G}_m$ -torsor over Z , and makes $L_Z = \mathcal{O}_Z(1)$ invertible, with $\mathcal{O}_Z(k) = L_Z^{\otimes k}$. A positive power of this line bundle is ample by the construction of Proj, so L_Z is ample.

The same charts show that Z is reduced, equidimensional, and Cohen–Macaulay, of dimension $N - 1$. The Hilbert leading coefficient gives $L_Z^{N-1} = 1$. Each irreducible projective component has a positive integral degree for this actual ample line bundle. There is therefore only one component, and reducedness makes Z integral. The image V_0 of E_1 in $H^0(Z, L_Z)$ has dimension N : restriction is injective, since a nonzero homogeneous element in a reduced positive-dimensional equidimensional ring cannot vanish on its whole puncture.

We construct successive Cartier sections while tracking the images of V_0 , not assuming surjectivity onto full spaces of sections. Suppose Z_i is a positive-dimensional integral projective Cohen–Macaulay scheme obtained after i cuts, and that the image V_i of V_0 on it has dimension $N - i$. Choose a nonzero section $s_i \in V_i$. Its zero scheme Z_{i+1} is an effective Cartier divisor. It is nonempty, because $L_Z|_{Z_i}$ is ample of positive degree, and it is pure and Cohen–Macaulay. Its degree is again one. The component-degree formula forces a single component of generic multiplicity one; the S_1 property then makes this divisor reduced, hence integral. The kernel of restriction $V_i \rightarrow H^0(Z_{i+1}, L_Z|_{Z_{i+1}})$ is exactly $\mathbb{C}s_i$: division by s_i identifies that kernel with regular global functions on the proper integral scheme Z_i , and those functions are constant. Thus $\dim V_{i+1} = N - i - 1$.

After $N - 1$ cuts, the resulting scheme is a single reduced point and the remaining image space has dimension one. Choose a final section nonzero at that point. Lifting all these sections to E_1 gives N elements whose common zero on $\text{Proj } E$ is empty. Their ideal is therefore irrelevant-primary and they form a homogeneous system of parameters. Cohen–Macaulayness makes the sequence regular. The quotient has Hilbert series

$$(1 - z)^N \text{Hilb } E(z) = 1.$$

It is $\mathbb{C}$, so induction on degree, or graded Nakayama, shows that these N linear elements generate E . The resulting surjection from a polynomial ring on N degree-one variables is an isomorphism by the same Hilbert series. $\square$

Corollary 6.6 (A nontrivial second stabilizer). *There is a nonvertex closed point of $\text{Spec } D$ whose T -grading stabilizer has order $h \geq 2$.*

Proof. Otherwise Lemma 6.5 would make D a polynomial ring on N degree-one generators, and Proposition 6.4 would make G a polynomial ring on $n = N + 1$ generators. Choose these generators homogeneous also for original degree and lift them to S . Terminating cancellation in the W -filtration shows that the lifts generate S . A surjection from a polynomial ring on n variables to the n -dimensional domain S has zero kernel. Thus C would be smooth, contrary to our standing hypothesis. Stabilizers at nonvertex points of a positive grading are finite, so the asserted order $h \geq 2$ exists. $\square$

7 The slice estimate and the supremum bootstrap

Throughout this section, $n = N + 1 \geq 5$, and the local theorem is assumed in dimensions smaller than n . Recall that $p_n = 2(N/n)^n$ and that M_n is the supremum of $d(x, X)$ over all singular boundary-zero klt germs of dimension n . The smooth-value upper bound gives $M_n \leq 1$; no bound by p_n is assumed. We first work with a fixed singular cone $o \in C = \text{Spec } S$ obtained from Proposition 2.5, such that $d(o, C) \geq p_n$. Its minimizing Reeb vector ξ is normalized by $A_C(\text{wt}_\xi) = n$.

Choose a sufficiently fast grading from Lemma 2.6 in the large-stabilizer range, and take $m = m_{\max} \geq r/N$. All constructions in Proposition 6.4 then apply. We shall use, in particular,

$$\bar{S} = \text{gr}_T S, \quad D = \bar{S}/(\bar{v}), \quad G = \text{gr}_W S = D[\mathbf{v}], \quad \text{Hilb}_G(z) = (1 - z)^{-n}. \quad (7.1)$$

Here $W(\mathbf{v}) = 1$, $W|_D = T$, and D has a positive T -grading. The Rees total space $\mathfrak{X} = \text{Spec Rees}_W S$ is normal and Gorenstein. Its parameter λ has action weight -1 , and the action is trivial on S . The central ring G has canonical filtration shift n . We also use the family $\text{Spec } \bar{S} \rightarrow \mathbb{A}_t^1$ defined by $t = \bar{v}$, whose parameter has T -degree zero. Away from $t = 0$ it has the graded description

$$\bar{S}[\bar{v}^{-1}] \simeq \mathbb{C}[\bar{v}, \bar{v}^{-1}][u_1, \dots, u_N], \quad T(\bar{v}) = 0, \quad T(u_i) = 1. \quad (7.2)$$

The equivariance in Equation (7.2) will be essential below.

7.1 A slice through a point with nontrivial stabilizer

By Corollary 6.6, there is a nonvertex closed point $y_D \in \text{Spec } D$ whose T -grading stabilizer is μ_h , with $h \geq 2$. Set $y = (y_D, 0) \in \text{Spec } D[\mathbf{v}]$, considered as a point of the central fiber of $\mathfrak{X}$. Its stabilizer for the Rees action is the same μ_h .

Lemma 7.1 (The uniformizing slice and its tests). *There is a normal Gorenstein algebraic germ $y \in Y$ of dimension n , preserved by μ_h , with a nonzero function $\lambda_Y \in \mathfrak{m}_{Y,y}$ of character -1 , such that Y is klt away from $\lambda_Y = 0$. Every quasi-monomial valuation u centered at y gives a quasi-monomial valuation w centered at $o \in C$. Writing $\ell = u(\lambda_Y) > 0$, one has*

$$A_C(w) \leq N\ell + A_Y(u), \quad \text{vol}(w) \leq \ell^{-n}. \quad (7.3)$$

The comparison is valid before klt at y has been established.

Proof. Choose a positive- W -degree homogeneous function f on $\mathfrak{X}$ with $f(y) \neq 0$. Such a function exists because y_D is not the vertex of the positively graded ring D ; lift a nonvanishing homogeneous function from the central fiber. Write $d = W(f) > 0$ and $c = f(y)$. On the open set $U = \{f \neq 0\} \subset \mathfrak{X}$, take the level slice $Y_0 = \{f = c\}$. The action map

$$\theta : Y_0 \times \mathbb{G}_m \longrightarrow U, \quad (q, a) \longmapsto a \cdot q,$$

is finite étale: it is the cover obtained by adjoining a root $a^d = f/c$ of an invertible function. We take the germ of Y_0 at y and denote it by Y . Normality and the Gorenstein property descend from the étale product description, and $\dim Y = \dim \mathfrak{X} - 1 = n$. The subgroup μ_h fixes y and preserves the level slice, since h divides d .

The restriction λ_Y of λ has character -1 . It is a nonzero divisor on Y : its pullback to $Y \times \mathbb{G}_m$ differs from the pullback of λ only by the invertible group character a^{-1} . It vanishes at y . Since $\mathfrak{X}_{\lambda \neq 0} \simeq C \times \mathbb{G}_m$ is klt, the étale product description also shows that Y is klt away from $\lambda_Y = 0$.

We record the function-field relation to C . In $K(C)(\lambda)$ the action fixes $K(C)$ and gives λ weight -1 , so

$$f = \lambda^{-d}s \quad \text{for some } s \in S, \quad \lambda_Y^d = s/c \quad \text{on } Y.$$

After inverting λ_Y , each component of the slice is finite over the open set $\{s \neq 0\} \subset C$. It dominates this open set: the equation is monic in λ_Y , and the resulting finite algebra is flat over $S[s^{-1}]$. Moreover λ_Y does not vanish identically on any component through y . Thus $K(Y)$ is a finite extension of $K(C)$, and S injects into $K(Y)$.

Extend u to $K(Y)(a)$ by the Gauss rule with a of value zero and residue transcendental over the residue field of u . This is the extension trivial on the group factor. Denote by V its restriction through θ to $K(\mathfrak{X})$. Its center is the generic point of the orbit through y , and

$$A_{\mathfrak{X}}(V) = A_Y(u), \quad V(\lambda) = \ell. \quad (7.4)$$

Indeed, on the product a log smooth model defining u may simply be multiplied by $\mathbb{G}_m$; the group parameter contributes no discrepancy. Pulling back log smooth models by the finite étale chart preserves the canonical divisor and discrepancy in the restricted-value scaling. These assertions do not require u to be μ_h -invariant.

Let $w = V|_{K(C)}$. Since elements of S are invariant under the Rees action, their pullbacks to $Y \times \mathbb{G}_m$ are independent of a , and

$$w(s') = u(s'|_Y) \quad (s' \in S). \quad (7.5)$$

Equivalently, w is the restriction of u along the finite extension $K(Y)/K(C)$. The Abhyankar equality is preserved by finite extension and restriction, so w is quasi-monomial [18, Proposition 3.7].

Write $F^p S = \{W \geq p\}$ for the second filtration. If $s' \in F^p S$, the Rees function $\lambda^{-p}s'$ is regular on $\mathfrak{X}$, hence its restriction is regular at y . Therefore

$$w(s') \geq p\ell. \quad (7.6)$$

Since $F^1 S$ is the vertex maximal ideal, this proves that w is centered exactly at o . Applying Lemma 6.2 with canonical shift $Q = n$, and then Equation (7.4), gives the discrepancy inequality in Equation (7.3).

Finally, Equation (7.1) implies

$$\dim_{\mathbb{C}} S/F^p S = \frac{p^n}{n!} + O(p^{n-1}). \quad (7.7)$$

For $p = \lceil k/\ell \rceil$, Equation (7.6) gives $F^p S \subseteq \mathfrak{a}_k(w)$. The valuation ideal has support only at o , so its colength is the same before and after localizing S at the vertex. Taking leading coefficients in Equation (7.7) proves $\text{vol}(w) \leq \ell^{-n}$. $\square$

Lemma 7.2 (The slice is klt). *The germ $y \in Y$ is klt.*

Proof. If it were not klt, there would be a prime divisor E on a log resolution of Y with $A_Y(E) \leq 0$ and center containing y . By Lemma 7.1, its center is contained in $\lambda_Y = 0$, so $a = \text{ord}_E(\lambda_Y) > 0$. Choose a log resolution also resolving this function and the ideal of y , and choose a closed point q of E over y . In local regular parameters at q , give a parameter defining E cost 1 and the other parameters positive costs tending to zero. The resulting quasi-monomial valuations u_ε , centered exactly at y , satisfy

$$A_Y(u_\varepsilon) = A_Y(E) + O(\varepsilon), \quad \ell_\varepsilon = u_\varepsilon(\lambda_Y) = a + O(\varepsilon).$$

Their restrictions w_ε are centered at o . Since C is klt, $A_C(w_\varepsilon) > 0$; hence Equation (7.3) either already gives a contradiction through a nonpositive upper bound for this discrepancy, or gives

$$2N^n \leq \widehat{\text{vol}}(o, C) \leq A_C(w_\varepsilon)^n \text{vol}(w_\varepsilon) \leq \left(N + \frac{A_Y(u_\varepsilon)}{\ell_\varepsilon}\right)^n.$$

The quantity inside the last power is positive whenever this chain is applicable, and its limit superior is at most N . Letting ε tend to zero contradicts $2N^n > N^n$. $\square$

Lemma 7.3 (Singularity of the slice). *The germ $y \in Y$ is singular.*

Proof. Suppose that Y were smooth at y . The étale product chart then implies that $\mathfrak{X}$ is smooth at the corresponding point. In its regular local ring, λ is an eigenfunction of character -1 for μ_h . Lift the product parameter $\mathbf{v} \in G = D[\mathbf{v}]$ equivariantly to a function $\tilde{v}$ of character 1. Its differential is nonzero: the class of $\mathbf{v}$ is nonzero in the cotangent space of $\text{Spec } G$ at $(y_D, 0)$, which is a quotient of the cotangent space of $\mathfrak{X}$.

Consequently $H = \{\tilde{v} = 0\} \subset \mathfrak{X}$ is smooth at y , and the germ of $\text{Spec } D$ at y_D is the hypersurface cut out in H by $\lambda|_H$. This is a nonzero equation: λ and $\tilde{v}$ form a regular sequence because $\mathbf{v}$ is regular in G . If $\lambda|_H$ has a nonzero linear part, D is smooth at y_D . Otherwise it is a minimal hypersurface equation whose character is $-1 \pmod{h}$, and this character is nontrivial.

We now use the other deformation $\bar{S}$ over $t = \bar{v}$. Its special fiber is D , and t is invariant under the T -action and therefore under μ_h . Complete its local ring at the point y_D of the special fiber. The completion is flat over $\mathbb{C}[[t]]$. Choose a minimal system of eigen-coordinates of the completed local ring of D and lift them equivariantly. Complete Nakayama gives a surjection

$$P = \mathbb{C}[[t, x_1, \dots, x_e]] \longrightarrow \hat{\mathcal{O}}_{\text{Spec } \bar{S}, y_D}$$

with kernel I , where t is fixed and every x_i is an eigen-coordinate. Flatness identifies I/tI with the kernel of the special-fiber presentation.

If D is smooth at y_D , this kernel is zero and Nakayama gives $I = 0$. In the other case it is generated by the minimal hypersurface relation of character -1 . Lift that generator to I and average it to its character to obtain a relation $\tilde{F}$ of character -1 . Nakayama gives $I = (\tilde{F})$. Because the base parameter is invariant and -1 is a nontrivial character, one has

$$\tilde{F}(t, 0, \dots, 0) = 0.$$

In both cases, substituting $x_1 = \dots = x_e = 0$ defines a μ_h -fixed formal section over $\mathbb{C}[[t]]$ through y_D .

Since y_D is not the vertex, there is a positive- T -degree homogeneous element of D nonzero at y_D . Lift it homogeneously to $\bar{S}$. Its value on the formal section is a unit, so remains nonzero after passing to $\mathbb{C}((t))$. On the other hand, Equation (7.2) identifies this generic fiber, with its μ_h -action, with standard affine N -space: each coordinate has character 1. For $h \geq 2$ its fixed-point scheme is exactly the origin, where every positive- T -degree function vanishes. This is the required contradiction. $\square$

7.2 The character contribution to colength

By Lemma 7.2, a minimizing valuation u_* exists at $y \in Y$ and is quasi-monomial [52, Theorem 1.2(1)]. It is invariant under μ_h : this follows from uniqueness up to scaling and invariance of log discrepancy under automorphisms, or directly from [51, Theorem 1.1 and Corollary 1.2]. Write

$$A' = A_Y(u_*), \quad \ell = u_*(\lambda_Y) > 0, \quad \sigma = \text{vol}(u_*) > 0,$$

and let w be the valuation of C constructed in Lemma 7.1.

Lemma 7.4 (Equal leading character lengths). *Let $R = \mathcal{O}_{Y,y}$ and let $\mathfrak{a}_t = \{g \in R : u_*(g) \geq t\}$. For every character $\chi \in \mathbb{Z}/h\mathbb{Z}$,*

$$\dim_{\mathbb{C}}(R/\mathfrak{a}_t)_{\chi} = \frac{\sigma}{h n!} t^n + o(t^n) \quad (t \rightarrow \infty).$$

Proof. The valuation ideals are μ_h -invariant, so these character quotients are defined. Multiplication by λ_Y^j gives an injection

$$(R/\mathfrak{a}_{t-j\ell})_{\chi} \hookrightarrow (R/\mathfrak{a}_t)_{\chi-j} \quad (j \geq 0).$$

Indeed, the value of a product is the sum of the values, so the kernel is exactly the indicated valuation cutoff. For any two characters one may choose $0 \leq j \leq h-1$ connecting them. With $H_0 = (h-1)\ell$, this gives, for each fixed χ ,

$$\text{length}(R/\mathfrak{a}_{t-H_0}) \leq h \dim_{\mathbb{C}}(R/\mathfrak{a}_t)_{\chi} \leq \text{length}(R/\mathfrak{a}_{t+H_0}).$$

The total length has asymptotic $\sigma t^n/n! + o(t^n)$, proving the assertion. The argument uses the primitive character of the nonzero function λ_Y , and requires no assumption on the tangent representation. $\square$

Proposition 7.5 (The refined slice estimate). *For the minimizing slice valuation and its restriction,*

$$\text{vol}(w) \leq \min_{0 \leq x \leq 1} \left\{ \frac{x^n}{\ell^n} + (1-x)^n \frac{\sigma}{h} \right\}. \quad (7.8)$$

Proof. Set $\mathfrak{a}_k = \{s \in S : w(s) \geq k\}$. For a positive integer p , the colength splits exactly as

$$\dim_{\mathbb{C}} S/\mathfrak{a}_k = \dim_{\mathbb{C}} S/(F^p S + \mathfrak{a}_k) + \dim_{\mathbb{C}} F^p S/(F^p S \cap \mathfrak{a}_k). \quad (7.9)$$

The first term is bounded by $\dim S/F^p S$. For the second, restriction and division define a linear map

$$F^p S \longrightarrow R, \quad s \longmapsto \lambda_Y^{-p}(s|_Y).$$

It is regular because $\lambda^{-p}s$ belongs to the Rees algebra. Its image has character $p \pmod{h}$, since s is invariant and λ_Y has character -1 . By Equation (7.5), it induces the injection

$$\frac{F^p S}{F^p S \cap \mathfrak{a}_k} \hookrightarrow (R/\mathfrak{a}_{k-p\ell})_{p \pmod{h}}. \quad (7.10)$$

In particular, this step is a comparison of one character quotient, not a comparison with the full length upstairs.

Fix $0 < x < 1$ and take $p = \lfloor kx/\ell \rfloor$. Apply Equation (7.7) to the first term in Equation (7.9), and Lemma 7.4 to Equation (7.10). There are only finitely many characters for the fixed slice, so the error is uniform as $p \pmod{h}$ varies. Multiplication by $n!/k^n$ and passage to the limit give

$$\text{vol}(w) \leq \frac{x^n}{\ell^n} + (1-x)^n \frac{\sigma}{h}.$$

Continuity extends this bound to the endpoints. Taking its minimum proves Equation (7.8). $\square$

Lemma 7.6 (Optimization). *For the fixed cone and large-stabilizer construction,*

$$d(o, C) \leq \frac{((N/n)\ell + A'/n)^n}{(\ell^{n/N} + (h/\sigma)^{1/N})^N} \quad (7.11)$$

$$\leq (N/n)^n + (A'/n)^n \sigma/h \leq \frac{p_n}{2} + \frac{M_n}{2}. \quad (7.12)$$

Equality in the middle, Hölder, inequality holds precisely when

$$(A')^N \ell \frac{\sigma}{h} = N^N. \quad (7.13)$$

Proof. For positive ℓ and σ , direct differentiation gives

$$\min_{0 \leq x \leq 1} \left\{ \ell^{-n} x^n + \frac{\sigma}{h} (1-x)^n \right\} = (\ell^{n/N} + (h/\sigma)^{1/N})^{-N}.$$

The minimum is attained at the unique interior point

$$x = \frac{\ell^{n/N}}{\ell^{n/N} + (h/\sigma)^{1/N}}. \quad (7.14)$$

Together with the discrepancy bound in Equation (7.3), this proves Equation (7.11).

Apply Hölder with conjugate exponents n/N and n to the vectors

$$(\ell, (h/\sigma)^{1/n}), \quad (N, A'(\sigma/h)^{1/n}).$$

Their scalar product is $N\ell + A'$. This gives

$$(N\ell + A')^n \leq (\ell^{n/N} + (h/\sigma)^{1/N})^N (N^n + (A')^n \sigma/h),$$

with equality exactly as in Equation (7.13). Finally, Lemmas 7.2 and 7.3 show that $y \in Y$ is a singular boundary-zero klt germ of dimension n . Since u_* computes its minimum,

$$(A'/n)^n \sigma = d(y, Y) \leq M_n.$$

Using $h \geq 2$ proves Equation (7.12). $\square$

7.3 The upper bound without a same-dimensional assumption

Theorem 7.7 (The upper bound). *Assume the local ODP volume theorem in all dimensions smaller than n . Then $M_n \leq p_n$. Equivalently, every singular boundary-zero klt n -dimensional germ satisfies*

$$\widehat{\text{vol}}(x, X) \leq 2(n-1)^n.$$

Proof. Suppose $M_n > p_n$. Choose singular klt germs whose densities $d_i > p_n$ tend to M_n . Proposition 2.5 gives for each of them a singular Gorenstein K-polystable cone $o_i \in C_i$, smooth off the vertex, with the same density d_i .

Fix i and choose the fast gradings of Lemma 2.6 for this cone. If $m_{\max} \leq r/N$ holds along an infinite subsequence, Theorem 3.1 implies that C_i is smooth or the standard ODP cone. The first possibility contradicts its singularity. The second contradicts $d_i > p_n$, since the normalized volume at an ODP is $2(n-1)^n$ by Lemma 2.3. Thus the large-stabilizer condition holds for all sufficiently far terms. Applying Lemma 7.6 to one such term gives

$$d_i \leq p_n/2 + M_n/2.$$

Letting i tend to infinity yields $M_n \leq p_n/2 + M_n/2$, contrary to $M_n > p_n$.

All choices of sufficiently fast gradings in this argument are made separately for each fixed cone C_i . Neither an attained supremum nor approximation estimates uniform over different cones are required. The slice bound used only the definition of M_n , so the argument assumes no same-dimensional ODP estimate. $\square$

7.4 Equality, a strict count, and the maximal stabilizer

Suppose now that the fixed cone satisfies $d(o, C) = p_n$. Theorem 7.7 is available. For every sufficiently fast grading in the large-stabilizer range, take $m = m_{\max}$ and make the construction above.

Lemma 7.8 (Consequences of equality). *One has $h = 2$, $d(y, Y) = p_n$, and $A' = N\ell$. After rescaling u_* so that $\ell = 1$, the resulting restriction w satisfies*

$$A' = N, \quad \sigma = 2, \quad A_C(w) = 2N, \quad \text{vol}(w) = 2^{-N}, \quad w = \frac{2N}{n} \text{wt}_\xi. \quad (7.15)$$

The optimal splitting fraction in Equation (7.8) is $x = 1/2$, and all inequalities producing that estimate are equalities at the level of leading coefficients.

Proof. The left endpoint of Equations (7.11)–(7.12) is now p_n , and the right endpoint is at most p_n . Every inequality is therefore an equality. In particular,

$$h = 2, \quad (A')^n \sigma = 2N^n.$$

Combining this with Equation (7.13) gives

$$N^N = (A')^N \ell \sigma / 2 = N^n \ell / A',$$

so $A' = N\ell$. Rescaling u_* by ℓ^{-1} makes $\ell = 1$, hence $A' = N$ and $\sigma = 2$. Equation (7.14) then gives $x = 1/2$ and the refined volume bound becomes $\text{vol}(w) \leq 2^{-N}$. The discrepancy bound becomes $A_C(w) \leq 2N$.

Since w is an admissible centered valuation,

$$2N^n = \widehat{\text{vol}}(o, C) \leq A_C(w)^n \text{vol}(w) \leq (2N)^n 2^{-N} = 2N^n.$$

Both factors are positive. Thus both displayed upper bounds are attained, and w computes the normalized volume minimum. Uniqueness up to scaling [51, Theorem 1.1], together with $A_C(\text{wt}_\xi) = n$, gives the last identity in Equation (7.15). $\square$

Lemma 7.9 (The distinguished section cannot have value greater than two). *In the normalization of Lemma 7.8, the section $v \in S_m$ used to define the two filtrations satisfies $w(v) \leq 2$.*

Proof. Suppose $w(v) > 2$. Choose a fixed real number t with

$$\frac{1}{w(v)} < t < \frac{1}{2},$$

and set $a_k = \lceil tk \rceil$ and $p_k = \lfloor k/2 \rfloor$. For large k one has $p_k - a_k > 0$. Use the vector-space basis $v^b e_i$ supplied by Lemma 6.1; it is adapted to W , with $W(v^b e_i) = b + T_i$. Let E_k be the span of its basis vectors with level less than $p_k - a_k$.

Multiplication by v^{a_k} maps E_k injectively into $S/F^{p_k}S$. To see injectivity modulo this filtration, the initial form of v is the regular polynomial variable $\mathbf{v}$ in $G = D[\mathbf{v}]$. Hence multiplication by v^{a_k} raises every nonzero initial level by exactly a_k and kills no initial form. Moreover, every element of $v^{a_k}E_k$ has w -value greater than k : w is nonnegative on S and $a_k w(v) > k$.

Thus the kernel of

$$S/F^{p_k}S \longrightarrow S/(F^{p_k}S + \mathfrak{a}_k)$$

has dimension at least

$$\dim E_k = \dim S/F^{p_k - a_k} S = \frac{(1/2 - t)^n}{n!} k^n + o(k^n).$$

This is a strict positive leading-order loss in the first term of the exact splitting Equation (7.9). The second term still satisfies the character estimate in Equation (7.10). At $x = 1/2$, $\ell = 1$, and $\sigma/h = 1$, these estimates give

$$\text{vol}(w) \leq 2^{-n} + 2^{-n} - (1/2 - t)^n < 2^{-N},$$

contradicting Lemma 7.8. Only the single exponent a_k is used at each cutoff, so there is no overlap between collections of powers to count. $\square$

Lemma 7.10 (Rounding the maximal stabilizer). *Along the fast gradings of the fixed equality cone, the large-stabilizer construction implies $Nm_{\max} \leq r$ for every sufficiently far term.*

Proof. Choose fixed torus eigen-generators $g_1, \dots, g_q$ of S , with characters $\chi_1, \dots, \chi_q$. Write

$$\zeta = \frac{r}{n}\xi + \varepsilon, \quad \|\varepsilon\| \rightarrow 0.$$

The ξ -weights of the generators are positive. Consequently there are fixed constants $c_1, c_2 > 0$ such that their ζ -degrees lie between $c_1 r$ and $c_2 r$ for all sufficiently far gradings. Also $m = m_{\max} \leq \max_i \deg_{\zeta} g_i = O(r)$, by Lemma 2.6.

The section $v \in S_m$ has a polynomial lift in these generators homogeneous for ζ . Every monomial in such a lift uses a bounded number of generators, because its degree is $m = O(r)$ and each generator degree is at least $c_1 r$. Its possible torus characters therefore belong to a fixed finite set, independent of the grading and of the chosen section v . If χ is any character occurring nontrivially in v , then

$$m = \langle \zeta, \chi \rangle = \frac{r}{n} \langle \xi, \chi \rangle + \langle \varepsilon, \chi \rangle,$$

and hence, uniformly for these characters,

$$\langle \xi, \chi \rangle = \frac{nm}{r} + O(\|\varepsilon\|/r).$$

Taking the minimum over the nonzero character components of v is precisely $\text{wt}_{\zeta}(v)$; the same estimate therefore holds even when v is not itself a torus eigenfunction. Lemma 7.8 gives

$$w(v) = \frac{2Nm}{r} + O(\|\varepsilon\|/r) = \frac{2Nm}{r} + o(1/r). \quad (7.16)$$

By Lemma 7.9, $w(v) \leq 2$. Thus $Nm - r \leq O(\|\varepsilon\|) = o(1)$. The left side is an integer, so it is nonpositive for all sufficiently far terms. On an exact rational ray the error is zero, and the same conclusion is immediate. All constants here concern the fixed cone C only. $\square$

Theorem 7.11 (The equality cone). *If a singular cone obtained in Proposition 2.5 satisfies $d(o, C) = p_n$, then it is the standard ordinary double point cone.*

Proof. Take the fast primitive gradings of Lemma 2.6. A term either already has $Nm_{\max} \leq r$, or lies in the large-stabilizer range. In the latter range, all constructions above are available for sufficiently far terms, and Lemma 7.10 again gives $Nm_{\max} \leq r$. The entire sequence therefore eventually satisfies the small-stabilizer hypothesis. Theorem 3.1 identifies C with the smooth affine cone or the standard quadratic cone. Since $o \in C$ is singular, only the ordinary double point remains. $\square$

8 Equality at the original germ and global consequences

We complete the return from the cone to the original singularity. The lci case of the gap theorem is already known [34, Theorem 1.3]. We include the hypersurface argument needed here, so that the source of strictness in the equality case is explicit.

8.1 A weighted hypersurface bound

Lemma 8.1 (A hypersurface test). *Let $x \in X$ be a klt hypersurface germ of dimension n , with equation F in regular ambient parameters $z_0, \dots, z_n$. For positive rational weights $w_0, \dots, w_n$, write d_F for the weighted order of F . Then*

$$\widehat{\text{vol}}(x, X) \leq \frac{d_F (\sum_{i=0}^n w_i - d_F)^n}{\prod_{i=0}^n w_i}. \quad (8.1)$$

Proof. Both sides of (8.1) are unchanged by a common scaling of the weights. We may therefore assume that they are positive integers. Choose a positive common multiple D_1 of the w_i , and in $\mathcal{O}_{X,x}$ put

$$I = (z_i^{D_1/w_i} : 0 \leq i \leq n).$$

This ideal is $\mathfrak{m}_x$ -primary.

The ambient weighted valuation has log discrepancy $\sum_i w_i$, gives value d_F to F , and value D_1 to the ambient ideal defining I . Hypersurface inversion of adjunction [11, Theorem 1.1], applied to the smooth ambient space with F of coefficient one, gives

$$\text{lct}_X(I) \leq \frac{\sum_i w_i - d_F}{D_1}. \quad (8.2)$$

In particular the numerator on the right is positive, because X is klt and I is primary.

Let J_t be the quotient weighted-order filtration on the hypersurface. The elementary monomial floor inequality gives

$$J_{(k+n+1)D_1} \subset I^k \subset J_{kD_1}.$$

Indeed, for a monomial of weight at least $(k+n+1)D_1$, the sum of the integers $\lfloor \alpha_i w_i / D_1 \rfloor$ is at least k . The associated graded of the quotient filtration is

$$\mathbb{C}[z_0, \dots, z_n] / (\text{in}_w F), \quad H(t) = \frac{1 - t^{d_F}}{\prod_i (1 - t^{w_i})}.$$

This formula holds even if $\text{in}_w F$ is reducible or nonreduced: it is a nonzero equation in a polynomial ring. The two filtration inclusions and the leading Hilbert coefficient give

$$e(I) = \frac{D_1^n d_F}{\prod_i w_i}. \quad (8.3)$$

Choose a divisor E computing $\text{lct}_X(I)$, and put $q = \text{ord}_E(I) > 0$. Its center is x , since I is primary. The inclusion $I^k \subset \mathfrak{a}_{kq}(E)$ implies

$$q^n \text{vol}(E) \leq e(I).$$

Testing the normalized-volume infimum with E , then using (8.2) and (8.3), yields

$$\widehat{\text{vol}}(x, X) \leq A_X(E)^n \text{vol}(E) \leq \text{lct}_X(I)^n e(I),$$

which is (8.1). □

Proposition 8.2 (Hypersurface equality). *If $x \in X$ is a singular klt hypersurface germ of dimension $n \geq 2$, then*

$$\widehat{\text{vol}}(x, X) \leq 2(n-1)^n.$$

If equality holds, the quadratic part of a local equation is nondegenerate, and (X, x) is analytically an ordinary double point.

Proof. Let $q = \text{mult}_x(F) \geq 2$. Weights all equal to one in Lemma 8.1 give

$$\widehat{\text{vol}}(x, X) \leq q(n+1-q)^n,$$

and klt implies $q < n+1$. The function $b(t) = t(n+1-t)^n$ satisfies

$$b'(t) = (n+1)(1-t)(n+1-t)^{n-1} < 0 \quad (1 < t < n+1).$$

Thus $q > 2$ gives a strict bound below $2(n-1)^n$.

Suppose $q = 2$ and the quadratic part is degenerate. After a linear change of coordinates, choose a direction absent from the quadratic part. Give it weight $1 - \epsilon$, and give every other direction weight one, where $0 < \epsilon < 1/3$ is rational. Every term of ordinary degree at least three has weight greater than two, so $d_F = 2$. The bound becomes

$$\frac{2(n-1-\epsilon)^n}{1-\epsilon}.$$

Its logarithmic derivative at $\epsilon = 0$ is $-n/(n-1) + 1 = -1/(n-1) < 0$. It is therefore strictly smaller than $2(n-1)^n$ for sufficiently small $\epsilon > 0$. Equality forces a nondegenerate quadratic part, and the holomorphic Morse Lemma gives the asserted analytic germ. $\square$

8.2 Completion of the induction

Proof of Theorem 1.1. The cases $n = 2, 3, 4$ were established in Section 2, the last by Theorem 2.4. Suppose the theorem is known in lower dimensions and let $n = N + 1 \geq 5$. The supremum argument of Theorem 7.7 gives $M_n \leq p_n$, hence the asserted inequality for every singular germ.

Now suppose $d(x, X) = p_n$. Apply Proposition 2.5 to obtain a K-polystable cone C with the same density and $\text{edim}(X, x) \leq \text{edim}(C, o)$. The equality argument of Theorem 7.11 makes C an ordinary double point. Therefore $\text{edim}(X, x) \leq n+1$.

Since x is singular, its embedding dimension is exactly $n+1$, and it is a hypersurface germ. To see this algebraically, start with an embedding in a smooth ambient variety and use equations with independent linear parts to cut the ambient germ down to dimension $n+1$. The remaining ideal is a height-one prime in a regular local ring, so is principal. Proposition 8.2 shows that its equation has nondegenerate quadratic part and hence that the analytic germ is ODP.

Conversely, Lemma 2.3 gives equality at every analytic ODP. This proves both directions of the equality statement and completes the induction. $\square$

8.3 The global comparison

Proof of Corollary 1.2. For every closed point x of a K-semistable $\mathbb{Q}$ -Fano n -fold V , the local-to-global comparison gives

$$(-K_V)^n \leq \left(\frac{n+1}{n}\right)^n \widehat{\text{vol}}(x, V).$$

This follows from [33, Theorem 21]; K-semistability and Ding semistability agree in this setting by [33, Theorem 13(2)]. If V is singular, apply Theorem 1.1 at a singular closed point:

$$(-K_V)^n \leq \left(\frac{n+1}{n}\right)^n 2(n-1)^n = 2 \left(\frac{n^2-1}{n}\right)^n < 2n^n.$$

If V is smooth and $V \not\cong \mathbb{P}^n$, the theorem of Li–Miao gives $(-K_V)^n \leq 2n^n$; its equality cases are a smooth quadric and $\mathbb{P}^1 \times \mathbb{P}^{n-1}$ [28, Theorem 1.1]. Both have Kähler–Einstein metrics, given by the homogeneous quadric metric and the suitably scaled Fubini–Study product metric, respectively. They are therefore K-polystable [3, Theorem 1.1]. Their degrees are

$$(-K_{Q^n})^n = (nH)^n = 2n^n, \quad (-K_{\mathbb{P}^1 \times \mathbb{P}^{n-1}})^n = (2H_1 + nH_2)^n = 2n^n.$$

For $n = 2$, a smooth quadric surface is $\mathbb{P}^1 \times \mathbb{P}^1$. The strict singular bound excludes every other equality case. $\square$

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